spin lecture

beann

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1 Introduction

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Dirac's Belt Trick, Topology, and Spin-$\frac{1}{2}$ Particles Lecture Notes on Quantum Spin

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2 Introduction: The Dirac Belt Trick

The **Dirac belt trick** is a striking physical demonstration invented by Paul Dirac to illustrate the bizarre rotational properties of spin-$\frac{1}{2}$ particles such as electrons, protons, and neutrons.

2.1 The Setup

Take a belt with one end fixed (e.g., held in place by a heavy book) and the other end free. The rules are:

[label=(i)]

1. You are *not* allowed to rotate either end of the belt.
2. You *are* allowed to translate (move) both ends through space without turning them.

2.2 The Observation

- 360° **twist:** Twist the free end by 360° (2π radians). No matter what manipulations you perform (without rotating the ends), *you cannot undo the twist.*
- 720° **twist:** Twist the free end by 720° (4π radians). Remarkably, the twist *can* be undone by looping the belt around, all without rotating either end.

This mirrors the behavior of spin-$\frac{1}{2}$ particles: a 2π rotation does *not* return the particle to its original quantum state, but a 4π rotation does.

3 The Space of Rotations: SO(3)

3.1 Specifying a Rotation

Any rotation in three-dimensional space is specified by two pieces of information:

1. An **axis of rotation** $\hat{n} \in R^3$, $|\hat{n}| = 1$.
2. An **angle of rotation** $\theta \in [0, \pi]$ (measured in radians, counterclockwise via the right-hand rule).

These can be packaged into a single vector $\vec{v} = \theta\,\hat{n}$, where the direction encodes the axis and the length encodes the angle.

3.2 Visualizing the Rotation Space

The space of all rotations can be visualized as a **solid ball of radius** π in R^3:

- Each point inside the ball corresponds to a unique rotation (given by the vector from the center to that point).
- The center of the ball corresponds to the identity rotation (no rotation).
- **Antipodal identification:** A rotation by π radians clockwise around an axis is the same as a rotation by π radians counterclockwise around the same axis. Therefore, *diametrically opposite points on the boundary of the ball must be identified as the same rotation.*

Remark 3.1 *Topologically,* SO(3) *is* not *simply a ball; it is a ball with antipodal boundary points identified. This identification is the key to understanding the belt trick.*

3.3 Rotation Matrices and SO(3)

Rotations act on position vectors $\vec{r} \in R^3$ via 3×3 orthogonal matrices. For example, a rotation about the x-axis by angle θ:

$$R_x(\theta) = (\,1\,)\,000\cos\theta - \sin\theta 0 \sin\theta \cos\theta.$$

Definition 3.1 (Orthogonal Matrix) *A matrix R is **orthogonal** if $R^\top R = I$, i.e., it preserves the inner product:*

$$\langle R\vec{r}_1,\, R\vec{r}_2\rangle = \langle \vec{r}_1,\, \vec{r}_2\rangle \quad \forall\, \vec{r}_1, \vec{r}_2 \in R^3.$$

Definition 3.2 (SO(3)) *The **special orthogonal group** SO(3) is the group of all 3×3 real orthogonal matrices with determinant $+1$:*

$$\mathrm{SO}(3) = \{R \in R^{3\times 3} : R^\top R = I,\ \det R = 1\}.$$

The "S" stands for "special" ($\det = 1$*), "O" for "orthogonal," and "3" for three dimensions.*

4 Paths, Belts, and Contractibility

4.1 Paths in the Space of Rotations

A **path** in SO(3) is a continuous curve $\gamma : [0, 1] \to \mathrm{SO}(3)$. A belt configuration corresponds to such a path: as you move along the belt from one end to the other, the local orientation of coordinate axes traces out a path in rotation space.

- A **perfectly straight belt** (no twist) corresponds to the **trivial path** — the path that stays at the identity (center of the ball).

- The constraint that neither end of the belt rotates means the path must be a **loop**: it starts and ends at the identity.
- "Undoing" a twist means **contracting** the loop to a point (the trivial path), without moving the endpoints.

4.2 Contractibility Results

Proposition 4.1 *The loop in* $\mathrm{SO}(3)$ *corresponding to a* 2π *twist is* ***not contractible*** *to the trivial path.*

Intuition: The 2π-twist loop passes through the boundary of the ball, connecting two antipodal points. Since these antipodal points are identified, the path forms a closed loop that "wraps around" the space in a way that cannot be undone.

Proposition 4.2 *The loop in* $\mathrm{SO}(3)$ *corresponding to a* 4π *twist* ***is contractible*** *to the trivial path.*

Intuition: The 4π-twist loop passes through the boundary twice. The two pairs of antipodal crossings can be brought together, allowing the path to be pulled through the boundary and contracted.

5 Groups

Definition 5.1 (Group) *A* ***group*** *is a set* G *equipped with a binary operation* $+$ *satisfying:*

[label=(1)]

1. ***Closure:*** $g_1, g_2 \in G g_1 + g_2 \in G$.
2. ***Identity:*** $\exists\, \mathrm{id} \in G$ *such that* $g + \mathrm{id} = g$ *for all* $g \in G$.
3. ***Associativity:*** $(g_1 + g_2) + g_3 = g_1 + (g_2 + g_3)$.
4. ***Inverses:*** *For each* $g \in G$, $\exists\, (-g) \in G$ *such that* $g + (-g) = \mathrm{id}$.

Example 5.1 (Rotations Form a Group) *The set* $\mathrm{SO}(3)$ *forms a group under composition of rotations:*

- *The "sum" of two rotations is performing one after the other.*
- *The identity element is the trivial rotation (do nothing).*
- *The inverse of a rotation by* θ *about* $\hat{n}$ *is a rotation by* θ *about* $-\hat{n}$.

6 The Fundamental Group of $\mathrm{SO}(3)$

6.1 Fundamental Group

Definition 6.1 (Fundamental Group $\pi_1(X)$) *Let X be a topological space with a chosen basepoint x_0. The* ***fundamental group*** *$\pi_1(X, x_0)$ is the set of all loops starting and ending at x_0, up to continuous deformation (homotopy), with the group operation being concatenation of loops.*

6.2 $\pi_1(\mathrm{SO}(3)) = Z/2Z$

In SO(3), with the basepoint at the identity:

- The **trivial loop** [id]: the path that stays at the identity.
- The **non-trivial loop** $[p]$: the path corresponding to the 2π twist, which is *not* contractible.

Adding $[p]$ to itself gives the 4π-twist loop, which *is* contractible:

$$[p] + [p] = [\mathrm{id}].$$

The full group table is:

| $+$ | [id] | | |

Theorem 6.1

$$\pi_1\big(\mathrm{SO}(3)\big) \cong Z/2Z \cong Z_2.$$

The Dirac belt trick is a physical demonstration of this theorem.

7 Quantum Spin States and SU(2)

7.1 Spin-$\frac{1}{2}$ States

Quantum spin states are vectors in C^2:

$$\psi = (\,\alpha\,)\,\beta, \quad \alpha, \beta \in C.$$

The inner product on C^2 involves the conjugate transpose:

$$\psi_1\psi_2 = \bar{\alpha}_1\alpha_2 + \bar{\beta}_1\beta_2.$$

Physical states satisfy the normalization condition $\psi\psi = |\alpha|^2 + |\beta|^2 = 1$.

7.2 Measurement Probabilities

If an electron has spin state $\psi = (\,\alpha\,)$
β, then upon measuring spin along the z-axis: P(spin up) $= —\alpha|^2 = \big|\uparrow_z\psi\big|^2$, $P(spindown) = |\beta|^2 = \big|\downarrow_z\psi\big|^2$.

The spin-up and spin-down states along each axis are:

$$\uparrow_x = \frac{1}{\sqrt{2}}\,(\,1\,)\,1, \quad \downarrow_x = \frac{1}{\sqrt{2}}\,(\,1\,)\,-1,$$

$$\uparrow_y = \frac{1}{\sqrt{2}}\,(\,1\,)\,i, \quad \downarrow_y = \frac{1}{\sqrt{2}}\,(\,1\,)\,-i,$$

$$\uparrow_z = (\,1\,)\,0, \quad \downarrow_z = (\,0\,)\,1.$$

7.3 Rotations of Spin States: SU(2)

Rotations act on spin states via 2×2 unitary matrices U with $\det U = 1$.

Definition 7.1 (SU(2)) *The* ***special unitary group*** SU(2) *is*

$$\mathrm{SU}(2) = \{U \in C^{2\times 2} : U^\dagger U = I,\ \det U = 1\}.$$

A general element of SU(2) has the form

$$U = (\,a\,)\,b - \bar{b}\bar{a}, \quad |a|^2 + |b|^2 = 1.$$

Writing $a = X + iY$ and $b = Z + iW$, the constraint becomes $X^2 + Y^2 + Z^2 + W^2 = 1$, which is the equation for the **3-sphere** $S^3 \subset R^4$:

$$\mathrm{SU}(2) \cong S^3 \quad (astopologicalspaces).$$

8 SU(2) as a Double Cover of SO(3)

8.1 Covering Spaces

Definition 8.1 (Covering Space) *A **covering space** of a topological space X is a space $\tilde{X}$ together with a continuous surjective map $p : \tilde{X} \to X$ such that every point of X has a neighborhood evenly covered by p. If each fiber $p^{-1}(x)$ has exactly two elements, we call it a **double cover**.*

8.2 The Double Cover SU(2) → SO(3)

The key topological relationship is:

Theorem 8.1 SU(2) *is a double cover of* SO(3)*: there is a* 2*-to-*1 *surjective group homomorphism*

$$\phi : \mathrm{SU}(2) \longrightarrow \mathrm{SO}(3).$$

The kernel is $\ker \phi = \{I, -I\}$*, so U and $-U$ map to the same rotation.*

The covering map is constructed using the Pauli matrices. Given $\vec{r} = (x, y, z)$, form the traceless Hermitian matrix

$$\vec{r} \cdot \vec{\sigma} = x\sigma_x + y\sigma_y + z\sigma_z = \begin{pmatrix} z & x - iy \\ x + iy & -z \end{pmatrix}.$$

Under conjugation by $U \in \mathrm{SU}(2)$:

$$U(\vec{r} \cdot \vec{\sigma})U^\dagger = \vec{r}' \cdot \vec{\sigma},$$

where $\vec{r}' = R\vec{r}$ for some $R \in \mathrm{SO}(3)$. This defines the map $\phi(U) = R$.

8.3 Lifting Paths and Non-Contractibility

When we lift the 2π-twist loop from SO(3) to SU(2):

- The lifted path starts at $I \in \mathrm{SU}(2)$ but ends at $-I \in \mathrm{SU}(2)$.
- Since the endpoints differ, the lifted path is not a loop, proving by contradiction that the original loop in SO(3) is non-contractible.

The 4π-twist loop lifts to a loop in SU(2) (starts and ends at I), which *is* contractible since $\pi_1(S^3) = \{e\}$ (the trivial group — every loop on S^3 is contractible).

8.4 Fundamental Group Relationship

$$\pi_1\big(\mathrm{SU}(2)\big) = \{e\}, \quad \pi_1\big(\mathrm{SO}(3)\big) = Z_2.$$

The fundamental group of SO(3) has exactly *twice* as many elements as that of SU(2), reflecting the *double* cover.

9 Why Electrons Negate Under 2π Rotation

The physical consequence of all the above mathematics:

1. The group that acts on C^2 spin vectors is SU(2), not SO(3).
2. To rotate an electron, we must **lift** a path from SO(3) to SU(2).
3. A 2π rotation, when lifted from $I \in \mathrm{SU}(2)$, ends at $-I \in \mathrm{SU}(2)$.
4. A 4π rotation returns to $I \in \mathrm{SU}(2)$.

Theorem 9.1 (Spin-$\frac{1}{2}$ Rotation Property) *For a spin-$\frac{1}{2}$ particle: Rotation by 2π : $\psi \mapsto -\psi$,*
Rotationby4π : $\psi \mapsto +\psi$.

The explicit SU(2) matrix for a rotation by angle θ about axis $\hat{n}$ is:

$$U(\hat{n}, \theta) = \cos\frac{\theta}{2} I - i \sin\frac{\theta}{2} (\hat{n} \cdot \vec{\sigma}).$$

Substituting $\theta = 2\pi$:

$$U(\hat{n}, 2\pi) = \cos\pi\, I - i \sin\pi\, (\hat{n} \cdot \vec{\sigma}) = -I.$$

This confirms that a 2π rotation negates the spin state.

10 Physical Realization and Detection

10.1 How to Rotate an Electron

1. **Change of reference frame:** Walking 2π radians around an electron while continuously describing its state in your own frame results in a negated state.
2. **Magnetic field (Larmor precession):** An electron's spin angular momentum vector precesses around an external magnetic field. By tuning the field strength and duration, any desired rotation can be achieved.

10.2 Detecting the Minus Sign

The overall phase -1 from a 2π rotation is *observable* via interference:

1. Place an electron (or neutron) in a quantum superposition at two spatial locations.
2. Subject one copy to a magnetic field inducing a 2π rotation (introducing the minus sign).
3. Recombine the two copies.

Interference effects from the relative minus sign are measurable. This has been confirmed experimentally via **neutron interferometry**.

11 Additional Topological Remarks

11.1 Real Projective Space

Identifying antipodal points on the 2-sphere S^2 yields 2-dimensional real projective space:

$$RP^2 = S^2/(x \sim -x).$$

Extending to the 3-sphere:

$$RP^3 = S^3/(x \sim -x).$$

Topologically, SO(3) $\cong RP^3$, which is consistent with SU(2) $\cong S^3$ being a double cover.

11.2 Nature and Spin

The fundamental group $\pi_1(\mathrm{SO}(3)) = Z_2$ constrains what kinds of particles nature can produce. A spin-$\frac{1}{3}$ particle is impossible because $Z_2 \neq Z_3$. Nature's creativity operates within the rigid bounds of mathematical structure.

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Quantum Spin (1): Introduction Lecture Notes on Quantum Spin

Contents

12 Motivation

Quantum spin for spin-$\frac{1}{2}$ particles (electrons, protons, neutrons) is often described through analogies: "spin is *like* the particle is actually spinning, but not really," or "spin is *like* an angular momentum vector, but not really." These analogies, while intuitive, obscure the mathematical simplicity of quantum spin.

The goal of these notes is to describe *what spin is* — not merely what it is like. The only prerequisite is familiarity with complex numbers.

13 Review of Complex Numbers

Definition 13.1 (Complex Numbers) *The set of* ***complex numbers*** *C consists of all numbers of the form $z = a + bi$, where $a, b \in R$ and $i^2 = -1$. Here $a = Re(z)$ is the* ***real part*** *and $b = Im(z)$ is the* ***imaginary part****.*

Definition 13.2 (Complex Conjugate) *If $z = a + bi$, the* ***complex conjugate*** *is $\bar{z} = a - bi$.*

Definition 13.3 (Magnitude Squared) *The* ***magnitude squared*** *of z is*

$$|z|^2 \bar{z}\, z = (a - bi)(a + bi) = a^2 + b^2.$$

Since $a, b \in R$, we have $|z|^2 \geq 0$, with equality if and only if $z = 0$.

14 Vectors in R^3: A Quick Review

14.1 Components and Basis Vectors

A familiar vector $\vec{L} \in R^3$ is specified by three real components:

$$\vec{L} = (\,L\,)_x\, L_y L_z = L_x\, \hat{\imath} + L_y\, \hat{\jmath} + L_z\, \hat{k},$$

where the standard basis vectors are

$$\hat{\imath} = (\,1\,)\,00, \quad \hat{\jmath} = (\,0\,)\,10, \quad \hat{k} = (\,0\,)\,01.$$

14.2 The Dot Product

The **dot product** of $\vec{v}, \vec{u} \in R^3$ is

$$\vec{v} \cdot \vec{u} = v_x u_x + v_y u_y + v_z u_z \in R.$$

Key properties:

- Symmetry: $\vec{v} \cdot \vec{u} = \vec{u} \cdot \vec{v}$.
- Basis orthonormality: $\hat{\imath} \cdot \hat{\imath} = 1$, $\hat{\imath} \cdot \hat{\jmath} = 0$, etc.
- Component extraction: $v_x = \vec{v} \cdot \hat{\imath}$, $v_y = \vec{v} \cdot \hat{\jmath}$, $v_z = \vec{v} \cdot \hat{k}$.
- Geometric meaning: $\vec{v} \cdot \vec{u} = |\vec{v}|\,|\vec{u}| \cos\theta$, measuring how much two vectors "overlap."

15 Quantum Spin Vectors

15.1 Two Key Differences from Classical Vectors

Quantum spin vectors differ from ordinary R^3 vectors in two fundamental ways:

[label=**1.**]

1. **Dimension:** Spin vectors are **2-dimensional**, not 3-dimensional.
2. **Components:** Spin vectors have **complex** components, not real.

That is, quantum spin states are elements of C^2, not R^3.

15.2 Notation: Kets and Basis States

In Dirac notation, the basis vectors for spin-$\frac{1}{2}$ are written as **kets**:

$$\uparrow = (\,1\,)\,0, \qquad \downarrow = (\,0\,)\,1,$$

read as "spin up" and "spin down." A general spin state is

$$\psi = \alpha\uparrow + \beta\downarrow = (\,\alpha\,)\,\beta \in C^2,$$

where $\alpha, \beta \in C$.

Remark 15.1 *Despite the unfamiliar notation, the mathematics is essentially the same as for ordinary vectors — just with two complex components instead of three real ones.*

16 The Inner Product on C^2

16.1 Definition

The analog of the dot product for spin vectors involves complex conjugation:

Definition 16.1 (Inner Product on C^2) *For $\psi_1 = (\,\alpha\,)_1$ β_1 and $\psi_2 = (\,\alpha\,)_2$ β_2, the **inner product** is*

$$\psi_2\psi_1 = \bar{\alpha}_2\,\alpha_1 + \bar{\beta}_2\,\beta_1 \in C.$$

16.2 Properties

- The inner product is a single complex number (not a vector).
- **Conjugate symmetry** (not full symmetry):

$$\psi_1\psi_2 = \overline{\psi_2\psi_1}.$$

- The inner product of a state with itself is real and non-negative:

$$\psi\psi = |\alpha|^2 + |\beta|^2 \geq 0.$$

17 Dual Vectors (Bras)

Definition 17.1 (Dual Vector / Bra) *Given a ket* $\psi = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$*, the* ***dual vector*** *(or* ***bra****) is obtained by complex conjugating the components and transposing to a row vector:*

$$\psi = \begin{pmatrix} \bar{\ } \end{pmatrix} \alpha \bar{\beta}.$$

The inner product can then be written as a matrix product:

$$\psi_2 \psi_1 = \begin{pmatrix} \bar{\ } \end{pmatrix} \alpha_2 \bar{\beta}_2 \begin{pmatrix} \alpha \end{pmatrix}_1 \beta_1 = \bar{\alpha}_2 \alpha_1 + \bar{\beta}_2 \beta_1.$$

Remark 17.1 (Origin of the Names) *This notation was invented by Paul Dirac. A ket* ψ *is a vector; a bra* ψ *is a dual vector. Placing them together,* $\psi_2 \psi_1$*, forms a* ***bra-ket*** *(bracket) — the inner product.*

In matrix language:

- A ket is a 2×1 matrix (column vector).
- A bra is a 1×2 matrix (row vector).
- Their product $\cdot\cdot$ is a 1×1 matrix (a number).

18 Measurements and Probability

18.1 The Experimental Setup

An experimenter can ask one question about an electron's spin: *Is the spin up or down with respect to a particular axis?*

- There are only **two possible outcomes**: spin up or spin down.
- A theorist can only predict the **probabilities** of each outcome — not which one will occur with certainty.

18.2 Probability Rules

Given a state $\psi = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, the measurement probabilities along the z-axis are: P(spin up along z) = $|\alpha|^2 = \left|\uparrow_z \psi\right|^2$,

$P(spin down along z) = |\beta|^2 = \left|\downarrow_z \psi\right|^2$.

Special cases:

- $\psi = \uparrow_z = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$: always measured spin up (100%).
- $\psi = \downarrow_z = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$: always measured spin down (100%).

18.3 Normalization

Since probabilities must sum to 1:

$$|\alpha|^2 + |\beta|^2 = 1 \quad \Longleftrightarrow \quad \psi\psi = 1.$$

A state satisfying this condition is called **properly normalized** or a **physical state**.

19 Spin States Along Other Axes

The states that are always spin up or down along each axis are:

Axis	$\uparrow$	$\downarrow$
x	$1\sqrt{2}\,(\,1\,)$	$1\sqrt{2}\,(\,1\,)$
1		

-1

y$1\sqrt{2}\,(\,1\,)$
i $1\sqrt{2}\,(\,1\,)$
$-\,i$

z $(\,1\,)$
0 $(\,0\,)$
1

The factors of $1/\sqrt{2}$ ensure normalization: $\psi\psi = 1$.

19.1 General Measurement Rule

For *any* axis $a \in \{x, y, z\}$:

$$P(spinupalonga) = \left|\uparrow_a \psi\right|^2, \qquad P(spindownalonga) = \left|\downarrow_a \psi\right|^2.$$

19.2 Orthogonality

Along every axis, the spin-up and spin-down states are orthogonal:

$$\uparrow_a \downarrow_a = 0 \quad fora \in \{x, y, z\}.$$

19.3 Context-Dependent Definite Spin

Whether a particle has a definite spin depends on which axis you measure. For example, $\uparrow_x$ has definite spin along x (always up), but along z it yields:

$$P(upalongz) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2} = 50\%.$$

We say $\uparrow_x$ is in a **superposition** of the z-axis eigenstates:

$$\uparrow_x = \frac{1}{\sqrt{2}}\uparrow_z + \frac{1}{\sqrt{2}}\downarrow_z.$$

20 Worked Example

Consider the state

$$\psi = \frac{3}{5}\uparrow_z + \frac{4i}{5}\downarrow_z = (\,3\,)\,/54i/5.$$

Normalization check:

$$\psi\psi = \left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = \frac{9}{25} + \frac{16}{25} = 1.$$

Probabilities along the z-axis:

$$P(\uparrow_z) = \left|\frac{3}{5}\right|^2 = \frac{9}{25} = 36\%, \qquad P(\downarrow_z) = \left|\frac{4}{5}\right|^2 = \frac{16}{25} = 64\%.$$

Probabilities along the x-axis: $\mathrm{P}(\uparrow_x) = |\uparrow_x\psi|^2 = \left|\frac{1}{\sqrt{2}}\left(\frac{3}{5} + \frac{4i}{5}\right)\right|^2 = \frac{1}{2}\left(\frac{9}{25} + \frac{16}{25}\right) = 50\%$,
$P(\downarrow_x) = 50\%$.

Probabilities along the y-axis: $\mathrm{P}(\uparrow_y) = |\uparrow_y\psi|^2 = \left|\frac{1}{\sqrt{2}}\left(\frac{3}{5} + \frac{4i}{5}\cdot(-i)\right)\right|^2 = \left|\frac{1}{\sqrt{2}}\cdot\frac{7}{5}\right|^2 = \frac{49}{50} = 98\%$,
$P(\downarrow_y) = \frac{1}{50} = 2\%$.

This state is nearly aligned with the $+y$ direction.

21 Quantum Measurement and Collapse

21.1 The Collapse Postulate

After a measurement is performed, the state **collapses** onto the measured outcome:

- If the initial state is $\psi = \alpha\uparrow + \beta\downarrow$ and spin up is measured, then the post-measurement state becomes $\uparrow$.
- Subsequent identical measurements will yield spin up with 100% probability.

21.2 Collapse Prevents State Determination

Collapse prevents an experimenter from determining the quantum state (α, β) of a single electron with arbitrary precision. Each measurement alters the state, making it impossible to reconstruct the original state from repeated measurements on the *same* particle.

21.3 Intermediate Measurements Disturb the State

Consider an electron initially in state $\uparrow_z$:

1. Measuring along z: 100% spin up.
2. Now insert an intermediate x-axis measurement (yielding up or down, each 50%), *then* measure along z again.
3. Result: 50% spin up, 50% spin down along z — the intermediate measurement has destroyed the original information.

22 Global Phase Invariance

If $|z| = 1$ (i.e., $z = e^{i\theta}$ for some $\theta \in R$), then the states ψ and $z\psi$ are **physically indistinguishable**: all measurement probabilities are identical.

Proposition 22.1 *For any $z \in C$ with $|z| = 1$ and any axis a:*

$$\left|\uparrow_a z\psi\right|^2 = |z|^2 \left|\uparrow_a \psi\right|^2 = \left|\uparrow_a \psi\right|^2.$$

The factor $z = e^{i\theta}$ is called an **overall (global) phase** and is unobservable.

23 Spin Quantum Number and Angular Momentum

The spin quantum number s of a particle determines the dimension of its state space:

$$\dim = 2s + 1.$$

s	$2s+1$	State space	Angular momentum values ($m_s\hbar$)
0	1	C^1	0
1/2	2	C^2	$-\hbar/2,\ +\hbar/2$
1	3	C^3	$-\hbar,\ 0,\ +\hbar$
3/2	4	C^4	$-3\hbar/2,\ -\hbar/2,\ +\hbar/2,\ +3\hbar/2$

Angular momentum is quantized in units of $\hbar$: consecutive values always differ by exactly $\hbar$.

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24 Motivation: Expectation Values

Recall that for a state $\psi = (\,\alpha\,)$
β, the probability of measuring spin up (assigned value $+1$) or spin down (assigned value -1) along the z-axis gives the **expectation value**:

$$\langle S_z \rangle = (+1)\,|\alpha|^2 + (-1)\,|\beta|^2 = |\alpha|^2 - |\beta|^2.$$

Computing expectation values along x or y requires re-expressing ψ in the eigenbasis of those axes, which is cumbersome. The Pauli matrices provide a far more elegant approach.

24.1 Expectation Value Along x by Decomposition

Writing $\uparrow_z$ and $\downarrow_z$ in the x-basis: $\uparrow_z = \frac{1}{\sqrt{2}}\uparrow_x + \frac{1}{\sqrt{2}}\downarrow_x$,
$\downarrow_z = \frac{1}{\sqrt{2}}\uparrow_x - \frac{1}{\sqrt{2}}\downarrow_x$.

For a general state $\psi = \alpha\uparrow_z + \beta\downarrow_z$, collecting terms in the x-basis:

$$\psi = \frac{\alpha+\beta}{\sqrt{2}}\uparrow_x + \frac{\alpha-\beta}{\sqrt{2}}\downarrow_x.$$

Therefore:

$$\langle S_x \rangle = \frac{1}{2}|\alpha+\beta|^2 - \frac{1}{2}|\alpha-\beta|^2.$$

This works but is tedious. The Pauli matrices offer a better way.

25 Constructing the Pauli Matrices

25.1 The Key Idea: Operators with Eigenvectors

We seek a linear operator $\sigma_x : C^2 \to C^2$ satisfying:

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1. **Linearity:** $\sigma_x(\psi_1 + \psi_2) = \sigma_x\psi_1 + \sigma_x\psi_2$.
2. **Eigenvalue equations:**

$$\sigma_x\uparrow_x = +1 \cdot \uparrow_x, \qquad \sigma_x\downarrow_x = -1 \cdot \downarrow_x.$$

If such an operator exists, then for any state $\psi = \alpha_x\uparrow_x + \beta_x\downarrow_x$:

$$\sigma_x\psi = \alpha_x\uparrow_x - \beta_x\downarrow_x,$$

and the inner product gives the expectation value directly:

$$\psi\sigma_x\psi = |\alpha_x|^2 - |\beta_x|^2 = \langle S_x \rangle.$$

25.2 Finding σ_x Explicitly

The operator σ_x swaps the two components of a vector:

$$\sigma_x \,(\,\alpha\,)\,\beta = (\,\beta\,)\,\alpha.$$

Verification of linearity:

$$\sigma_x\,((\,\alpha\,)_1\,\beta_1 + (\,\alpha\,)_2\,\beta_2) = (\,\beta\,)_1 + \beta_2\alpha_1 + \alpha_2 = \sigma_x\,(\,\alpha\,)_1\,\beta_1 + \sigma_x\,(\,\alpha\,)_2\,\beta_2.$$

Verification of eigenvalue equations: $\sigma_x{\uparrow}_x = \sigma_x \frac{1}{\sqrt{2}}\,(\,1\,)$
$1 = \frac{1}{\sqrt{2}}\,(\,1\,)$
$1 = +1 \cdot {\uparrow}_x.$
$\sigma_x{\downarrow}_x = \sigma_x \frac{1}{\sqrt{2}}\,(\,1\,)$
$-\,1 = \frac{1}{\sqrt{2}}\,(\,-\,)\,1$
$1 = -1 \cdot {\downarrow}_x.$

26 Matrices: Definition and Multiplication

26.1 What Is a Matrix?

A 2×2 **matrix** is a linear function that maps one column vector to another:

$$(\,a\,)\,bcd\,(\,\alpha\,)\,\beta = (\,a\,)\,\alpha + b\beta c\alpha + d\beta.$$

The operator σ_x can therefore be written as:

$$\sigma_x = (\,0\,)\,110$$

26.2 General Matrix Multiplication

If A is an $m \times n$ matrix and B is an $n \times \ell$ matrix, the product AB is an $m \times \ell$ matrix with entries:

$$(AB)_{ij} = \sum_{k=1}^{n} A_{ik}\,B_{kj}.$$

Important special cases:

- **Matrix $\times$ column vector:** An $m \times m$ matrix times an $m \times 1$ column vector yields an $m \times 1$ column vector.
- **Row vector $\times$ column vector (inner product):** A $1 \times m$ row vector times an $m \times 1$ column vector yields a 1×1 scalar.
- **Column vector $\times$ row vector (outer product):** An $m \times 1$ column vector times a $1 \times m$ row vector yields an $m \times m$ matrix.

26.3 Associativity

Matrix multiplication is **associative**: $(AB)C = A(BC)$. This is crucial for manipulating expressions like $\psi\sigma_x\psi$, which can be evaluated as either $(\psi)(\sigma_x\psi)$ or $(\psi\sigma_x)(\psi)$.

27 The Three Pauli Matrices

Definition 27.1 (Pauli Matrices) *The three **Pauli matrices** are:*

$$\sigma_x = (\,0\,)\,110, \qquad \sigma_y = (\,0\,) - ii0, \qquad \sigma_z = (\,1\,)\,00 - 1$$

27.1 Eigenvalue Verification

Each Pauli matrix has eigenvalues $+1$ and -1, with the corresponding spin-up and spin-down states along that axis as eigenvectors.

For σ_y: $\sigma_y{\uparrow}_y = (\,0\,) - i$
$i0\frac{1}{\sqrt{2}}\,(\,1\,)$
$i = \frac{1}{\sqrt{2}}\,(\,(\,) - i)(i)$
$(i)(1) = \frac{1}{\sqrt{2}}\,(\,1\,)$
$i = +1 \cdot {\uparrow}_y.$

$\sigma_y{\downarrow}_y = (\,0\,) - i$
$i0\frac{1}{\sqrt{2}}\,(\,1\,)$
$-\,i = \frac{1}{\sqrt{2}}\,(\,-\,)\,1$
$i = -1 \cdot \frac{1}{\sqrt{2}}\,(\,1\,)$
$-\,i = -1 \cdot {\downarrow}_y.$

For σ_z: Since σ_z is diagonal, it simply multiplies each component by its diagonal entry: $\sigma_z{\uparrow}_z = (\,1\,)\,0$
$0 - 1\,(\,1\,)$
$0 = (\,1\,)$
$0 = +1 \cdot {\uparrow}_z.$

$\sigma_z{\downarrow}_z = (\,1\,)\,0$
$0 - 1\,(\,0\,)$
$1 = (\,0\,)$
$-\,1 = -1 \cdot {\downarrow}_z.$

28 Expectation Values via Pauli Matrices

Theorem 28.1 (Expectation Value Formula) *For any normalized spin state $\psi = (\,\alpha\,)$
β and axis $a \in \{x, y, z\}$:*

$$\langle S_a \rangle = \psi\sigma_a\psi.$$

Explicitly: $\langle S_x \rangle = \psi \sigma_x \psi = \bar{\alpha}\beta + \bar{\beta}\alpha$,
$\langle S_y \rangle = \psi \sigma_y \psi = -i\bar{\alpha}\beta + i\bar{\beta}\alpha$,
$\langle S_z \rangle = \psi \sigma_z \psi = |\alpha|^2 - |\beta|^2$.

Remark 28.1 *One can verify that* $\bar{\alpha}\beta + \bar{\beta}\alpha = \frac{1}{2}|\alpha+\beta|^2 - \frac{1}{2}|\alpha-\beta|^2$*, confirming consistency with the decomposition method.*

29 Eigenvectors, Eigenvalues, and Terminology

Definition 29.1 (Eigenvector and Eigenvalue) *If A is a matrix and v is a nonzero vector such that*

$$Av = \lambda v$$

for some scalar λ, then v is an ***eigenvector*** *of A with* ***eigenvalue*** *λ.*

Remark 29.1 *The prefix "eigen" comes from the German word meaning "same" or "own." An eigenvector is a vector that the matrix "keeps the same" (up to scaling).*

Summary of eigenvalue structure:

Matrix	**Eigenvalue** $+1$	**Eigenvalue** -1
σ_x	$\uparrow_x = \frac{1}{\sqrt{2}}\begin{pmatrix}1\end{pmatrix}$	$\downarrow_x = \frac{1}{\sqrt{2}}\begin{pmatrix}1\end{pmatrix}$
1		

-1

σ_y $\uparrow_y = \frac{1}{\sqrt{2}}\begin{pmatrix}1\end{pmatrix}$
i $\downarrow_y = \frac{1}{\sqrt{2}}\begin{pmatrix}1\end{pmatrix}$
$-i$

σ_z $\uparrow_z = \begin{pmatrix}1\end{pmatrix}$
0 $\downarrow_z = \begin{pmatrix}0\end{pmatrix}$
1

30 Outer Product Decomposition

Each Pauli matrix can be expressed as a difference of outer products of its eigenstates:

$$\sigma_a = \uparrow_a\uparrow_a - \downarrow_a\downarrow_a, \quad a \in \{x, y, z\}.$$

Correspondingly, the identity matrix satisfies:

$$I = \uparrow_a\uparrow_a + \downarrow_a\downarrow_a,$$

which reflects the conservation of probability: for any axis, $P(\uparrow) + P(\downarrow) = 1$.

30.1 Explicit Verification for σ_x

$$\uparrow_x\uparrow_x = \frac{1}{2}\,(\,1\,)\,1\,(\,1\,)\,1 = \frac{1}{2}\,(\,1\,)\,111,$$

$$\downarrow_x\downarrow_x = \frac{1}{2}\,(\,1\,)\;-1\,(\,1\,)-1 = \frac{1}{2}\,(\,1\,)-1-11.$$

$$\uparrow_x\uparrow_x - \downarrow_x\downarrow_x = \frac{1}{2}\,(\,1\,)\,111 - \frac{1}{2}\,(\,1\,)-1-11 = (\,0\,)\,110 = \sigma_x.$$

31 Inner Products and Outer Products

Definition 31.1 (Inner Product vs. Outer Product) *For column vectors u and v in C^2:*

- ***Inner product:*** *$vu = vu$ is a 1×1 scalar (a number).*
- ***Outer product:*** *w is a 2×2 matrix.*

The naming is intuitive: in the inner product vu, the brackets face *inward*; in the outer product w, they face *outward*.

32 Spin Along an Arbitrary Direction

Given a unit vector $\hat{n} = (n_x, n_y, n_z) \in R^3$, the operator for spin measurement along $\hat{n}$ is:

$$\hat{n} \cdot \vec{\sigma} = n_x\,\sigma_x + n_y\,\sigma_y + n_z\,\sigma_z = (\,n\,)_z\,n_x - in_y n_x + in_y - n_z.$$

The state that is always spin up in the $\hat{n}$ direction is the eigenvector of $\hat{n} \cdot \vec{\sigma}$ with eigenvalue $+1$.

Remark 32.1 *This connection between arbitrary spatial directions and 2×2 matrices is the foundation of the **Bloch sphere**, which will be developed in the next lecture.*

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33 Overview

The **Bloch sphere** provides a powerful geometric correspondence between quantum spin-$\frac{1}{2}$ states (vectors in C^2) and unit vectors in ordinary three-dimensional space ($\hat{n} \in R^3$, $|\hat{n}| = 1$). This correspondence allows us to visualize any spin state as a point on the unit sphere.

The key idea: a quantum state ψ corresponds to the direction $\hat{n}$ such that measuring spin along $\hat{n}$ always yields spin up (100% probability).

34 Spherical Coordinate Parametrization

A unit vector $\hat{n} \in R^3$ can be parametrized by two angles:

- $\theta \in [0, \pi]$: the **polar angle** between $\hat{n}$ and the z-axis.
- $\phi \in [0, 2\pi)$: the **azimuthal angle** between the projection of $\hat{n}$ onto the xy-plane and the x-axis.

The Cartesian components are:

$$n_x = \sin\theta\cos\phi, \qquad n_y = \sin\theta\sin\phi, \qquad n_z = \cos\theta.$$

35 The Correspondence

35.1 From $\hat{n}$ to ψ: The Spin-Up State Along $\hat{n}$

Given a direction $\hat{n}$ specified by (θ, ϕ), the quantum state that is always measured spin up along $\hat{n}$ is:

$$\uparrow_{\hat{n}} = (\cos)\,\theta 2 e^{i\phi} \sin\theta 2.$$

The convention is that the first component is real and non-negative.

35.2 From ψ to $\hat{n}$: The Bloch Vector

Given a state ψ, the corresponding unit vector (the **Bloch vector**) is:

$$\hat{n} = (\ )\,\psi\sigma_x\psi\psi\sigma_y\psi\psi\sigma_z\psi = \psi\vec{\sigma}\psi,$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is the "vector of Pauli matrices."

Remark 35.1 (Phase Ambiguity) *States ψ and $e^{i\gamma}\psi$ (differing by an overall phase) correspond to the* same *Bloch vector $\hat{n}$, since all expectation values are unchanged.*

Definition 35.1 (Bloch Sphere) *Since $\hat{n}$ is a unit vector ($|\hat{n}| = 1$), all Bloch vectors lie on the unit sphere $n_x^2 + n_y^2 + n_z^2 = 1$. This sphere is the* ***Bloch sphere****.*

36 Eigenvector Characterization

36.1 Spin Along $\hat{n}$ via $\hat{n} \cdot \vec{\sigma}$

Recall that the operator for spin measurement along $\hat{n}$ is:

$$\hat{n} \cdot \vec{\sigma} = n_x \sigma_x + n_y \sigma_y + n_z \sigma_z = (\, n \,)_z \, n_x - i n_y n_x + i n_y - n_z.$$

Using $n_x + i n_y = e^{i\phi} \sin\theta$, this becomes:

$$\hat{n} \cdot \vec{\sigma} = (\, \cos \,) \, \theta e^{-i\phi} \sin\theta e^{i\phi} \sin\theta - \cos\theta.$$

Theorem 36.1 *The state $\uparrow_{\hat{n}}$ is the eigenvector of $\hat{n} \cdot \vec{\sigma}$ with eigenvalue $+1$.*

Multiply:

$$(\hat{n} \cdot \vec{\sigma}) \uparrow_{\hat{n}} = (\, \cos \,) \, \theta e^{-i\phi} \sin\theta e^{i\phi} \sin\theta - \cos\theta \, (\, \cos \,) \, \frac{\theta}{2} e^{i\phi} \sin\frac{\theta}{2}.$$

Top component:

$$\cos\theta \cos\frac{\theta}{2} + \sin\theta \sin\frac{\theta}{2} = \cos\left(\theta - \frac{\theta}{2}\right) = \cos\frac{\theta}{2}.$$

Bottom component:

$$e^{i\phi}\left(\sin\theta \cos\frac{\theta}{2} - \cos\theta \sin\frac{\theta}{2}\right) = e^{i\phi} \sin\left(\theta - \frac{\theta}{2}\right) = e^{i\phi} \sin\frac{\theta}{2}.$$

Therefore:

$$(\hat{n} \cdot \vec{\sigma}) \uparrow_{\hat{n}} = (\, \cos \,) \, \frac{\theta}{2} e^{i\phi} \sin\frac{\theta}{2} = +1 \cdot \uparrow_{\hat{n}}.$$

36.2 The Spin-Down State Along $\hat{n}$

The eigenvector of $\hat{n} \cdot \vec{\sigma}$ with eigenvalue -1 is:

$$\downarrow_{\hat{n}} = (\, \sin \,) \, \theta 2 - e^{i\phi} \cos\theta 2.$$

36.3 Key Properties

These two states satisfy all expected relations: $\uparrow_{\hat{n}} \uparrow_{\hat{n}} = 1, \downarrow_{\hat{n}} \downarrow_{\hat{n}} = 1$, $\uparrow_{\hat{n}} \downarrow_{\hat{n}} = 0$, and the outer product decomposition:

$$\hat{n} \cdot \vec{\sigma} = \uparrow_{\hat{n}} \uparrow_{\hat{n}} - \downarrow_{\hat{n}} \downarrow_{\hat{n}}.$$

37 Sanity Checks

37.1 $\hat{n}$ Along the x-Axis

If $\hat{n} = \hat{x}$, then $\theta = \pi/2$, $\phi = 0$:

$$\uparrow_{\hat{n}} = (\, \cos \,) \, \frac{\pi}{4} e^{i \cdot 0} \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} \, (\, 1 \,) \, 1 = \uparrow_x.$$

37.2 $\hat{n}$ Along the y-Axis

If $\hat{n} = \hat{y}$, then $\theta = \pi/2$, $\phi = \pi/2$:

$$\uparrow_{\hat{n}} = (\cos)\frac{\pi}{4}e^{i\pi/2}\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}(1)\,i = \uparrow_y.$$

37.3 $\hat{n}$ Along the z-Axis

If $\hat{n} = \hat{z}$, then $\theta = 0$ (ϕ is arbitrary):

$$\uparrow_{\hat{n}} = (\cos)\,0e^{i\phi}\sin 0 = (1)\,0 = \uparrow_z.$$

37.4 Spin Up in $-\hat{n}$ Equals Spin Down in $\hat{n}$

To negate $\hat{n}$, send $\theta \to \pi - \theta$ and $\phi \to \phi + \pi$. Then: $\uparrow_{-\hat{n}} = (\cos)\,\frac{\pi-\theta}{2}$
$e^{i(\phi+\pi)}\sin\frac{\pi-\theta}{2} = (\sin)\,\frac{\theta}{2}$
$-\,e^{i\phi}\cos\frac{\theta}{2} = \downarrow_{\hat{n}}$.

38 Every State Points in Some Direction

Proposition 38.1 *For any normalized state $\psi \in C^2$ with $\psi\psi = 1$, there exists a direction $\hat{n}$ such that ψ is the spin-up state along $\hat{n}$ (up to a global phase).*

[Sketch] Given $\psi = (\alpha)$
β, multiply by a global phase $e^{i\gamma}$ to make the first component real and non-negative: $\alpha' = |\alpha| \geq 0$. Write the second component as $e^{i\phi}\beta'$ where $\beta' \geq 0$ is real. The normalization $(\alpha')^2 + (\beta')^2 = 1$ means we can parametrize $\alpha' = \cos\frac{\theta}{2}$, $\beta' = \sin\frac{\theta}{2}$ for some $\theta \in [0, \pi]$. This recovers the standard form.

39 Computing the Bloch Vector

For the state $\psi = (\cos)\,\frac{\theta}{2}$
$e^{i\phi}\sin\frac{\theta}{2}$, we verify $\hat{n} = \psi\vec{\sigma}\psi$:

39.1 x-Component

$\psi\sigma_x\psi = (\cos)\,\frac{\theta}{2}e^{-i\phi}\sin\frac{\theta}{2}\,(e)^{i\phi}\sin\frac{\theta}{2}$
$\cos\frac{\theta}{2}$
$= e^{i\phi}\cos\frac{\theta}{2}\sin\frac{\theta}{2} + e^{-i\phi}\sin\frac{\theta}{2}\cos\frac{\theta}{2}$
$= 2\cos\phi\cos\frac{\theta}{2}\sin\frac{\theta}{2} = \sin\theta\cos\phi = n_x$.

39.2 y-Component

$$\psi\sigma_y\psi = i\big(-e^{i\phi} + e^{-i\phi}\big)\cos\frac{\theta}{2}\sin\frac{\theta}{2} = 2\sin\phi\cos\frac{\theta}{2}\sin\frac{\theta}{2} = \sin\theta\sin\phi = n_y.$$

39.3 z-Component

$$\psi\sigma_z\psi = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = \cos\theta = n_z.$$

39.4 Bloch Vector is a Unit Vector

A quick manipulation confirms $|\hat{n}| = 1$:

$$\hat{n}\cdot\hat{n} = \psi(\hat{n}\cdot\vec{\sigma})\psi = \psi\psi = 1,$$

using the fact that ψ is an eigenvector of $\hat{n}\cdot\vec{\sigma}$ with eigenvalue $+1$.

40 Measurement Probability Formula

Theorem 40.1 (Overlap Probability) *The probability that a state pointing along $\hat{n}$ is measured to be spin up along direction $\hat{m}$ is:*

$$P(\uparrow_{\hat{m}}) = \left|\uparrow_{\hat{m}}\uparrow_{\hat{n}}\right|^2 = \frac{1+\hat{n}\cdot\hat{m}}{2}.$$

40.1 Special Cases

Let ϑ be the angle between $\hat{n}$ and $\hat{m}$ (so $\hat{n}\cdot\hat{m} = \cos\vartheta$):

$\hat{n}\cdot\hat{m}$	$P(\uparrow_{\hat{m}})$	Interpretation
$+1$ (parallel)	1	Certain to be spin up
0 (perpendicular)	1/2	Equal probability
-1 (antiparallel)	0	Certain to be spin down

40.2 Derivation

For $\hat{m} = \hat{z}$ and general $\hat{n}$:

$$\uparrow_z\uparrow_{\hat{n}} = (\,1\,)\,0\,(\,\cos\,)\,\frac{\theta}{2}e^{i\phi}\sin\frac{\theta}{2} = \cos\frac{\theta}{2}.$$

$$P(\uparrow_z) = \cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2} = \frac{1+\hat{n}\cdot\hat{z}}{2}.$$

The general formula follows by replacing $\hat{z}$ with $\hat{m}$ and $\cos\theta$ with $\hat{n}\cdot\hat{m}$.

41 Summary

1. Every spin-$\frac{1}{2}$ state corresponds to a point on the **Bloch sphere** — a unit sphere in R^3.

2. The **Bloch vector** $\hat{n} = \psi\vec{\sigma}\psi$ identifies the direction along which ψ is always spin up.

3. The **spin-up state** along $\hat{n} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta)$ is:

$$\uparrow_{\hat{n}} = (\cos)\frac{\theta}{2}e^{i\phi}\sin\frac{\theta}{2}.$$

4. Measurement probabilities are governed by $P(\uparrow_{\hat{m}}) = \frac{1+\hat{n}\cdot\hat{m}}{2}$.

5. The Bloch sphere makes the geometry of quantum spin as concrete as a classical direction vector.

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Quantum Spin (4): Classical Dynamics in a Magnetic Field & Motivating the Schrödinger Equation Lecture Notes on Quantum Spin

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42 The Central Questions of Physics

All of physics can be reduced to two questions:

1. **What are the physical states?** For quantum spin: two-dimensional complex vectors $\psi \in C^2$.
2. **How do they evolve in time?** This is what we now address.

The answer to the second question: a **magnetic field** causes a quantum spin state to change in time. Before treating this quantum mechanically, we build intuition from classical physics.

42.1 Setup

Throughout these notes, the physical situation is:

- An external magnetic field $\vec{B}$ permeates the region.
- A point-like electron with spin state $\psi(t)$ sits in this field.
- No external electric field ($\vec{E}_{ext} = 0$), so the electron remains stationary.
- We study only how the *spin state* evolves — not the spatial motion.

43 Classical Model: The Charged Spinning Sphere

43.1 The Model

We model the electron as a charged spinning sphere of uniform charge density, spinning with angular frequency $\vec{\Omega}_{spin}$. The angular momentum $\vec{L}$ points along the spin axis.

43.2 Forces from the Magnetic Field

The **Lorentz force law** for a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ is:

$$\vec{F} = q\,\vec{v} \times \vec{B}.$$

For a spinning charged sphere in a uniform $\vec{B}$:

- Different parts of the sphere move with different velocities.
- The forces on opposite sides are **equal and opposite** $\Rightarrow$ **no net force** (center of mass stays put).
- However, the forces create a **net torque** $\vec{\tau}$ on the sphere.

43.3 Torque Calculation

Recall $\vec{\tau} = \vec{r} \times \vec{F}$. For the sphere:

- On the left half: $\vec{r} \times \vec{F}$ points into the screen.
- On the right half: $\vec{r} \times \vec{F}$ *also* points into the screen.
- The torques **add** (unlike the forces, which cancel).

44 Larmor Precession

44.1 The Physics

Since $\vec{\tau} = \frac{d\vec{L}}{dt}$ and $\vec{\tau} \perp \vec{L}$, the angular momentum vector $\vec{L}$ traces out a circle around the magnetic field $\vec{B}$. This is **Larmor precession**: the angular momentum of a spinning charge precesses around the external magnetic field.

44.2 Classical Complications: Wobbling

In the classical case, the total angular velocity is $\vec{\Omega}_{total} = \vec{\Omega}_{spin} + \vec{\Omega}_{precession}$, introducing wobbling. However, if $\Omega_{spin} \gg \Omega_{precession}$, the wobbling is negligible and Larmor precession is the full story.

44.3 Quantum Simplification

In quantum mechanics, there is *no wobbling* — the spin state undergoes pure Larmor precession around $\vec{B}$. Heuristically, one can think of the quantum electron as "spinning infinitely fast," so the precession angular momentum is negligible.

45 Torque on a Current Loop

45.1 Setup: Square Current Loop

Consider a square loop of wire with side length L, carrying a total charge Q circulating with velocity v, placed in a uniform external field $\vec{B} = (B_x, B_y, B_z)$.

The current in the loop is:

$$I = \frac{Qv}{4L},$$

since the period of one full circuit is $T = 4L/v$.

45.2 Computing the Torque

Using $\vec{F} = q\vec{v} \times \vec{B}$ and $\vec{\tau} = \vec{r} \times \vec{F}$ for each segment, and summing over all four sides, the total torque is:

$$\vec{\tau}_{total} = \frac{QvL}{4}\,(-)\,B_y B_x 0 = IA\,(-)\,B_y B_x 0,$$

where $A = L^2$ is the area of the loop and $I = Qv/(4L)$ is the current.

45.3 The Magnetic Dipole Moment

Definition 45.1 (Magnetic Dipole Moment) *The **magnetic dipole moment** of a current loop is:*

$$\vec{\mu} = I\,\vec{A},$$

where $\vec{A}$ is the area vector (magnitude = area, direction = normal to the loop via the right-hand rule).

The torque takes the elegant form:

$$\vec{\tau} = \vec{\mu} \times \vec{B}.$$

Remark 45.1 *This formula is valid for* any *shape of current loop, not just squares. Any planar loop can be decomposed into a grid of tiny squares; internal currents cancel, and the areas and torques add up.*

46 Energy of a Current in a Magnetic Field

46.1 Work Done by the Field

Consider the square loop tilted at angle θ to $\vec{B}$. The magnetic force on each side is constant and horizontal (independent of θ), with magnitude $F = \frac{Q}{4}vB$.

When the loop rotates from angle θ_1 to θ_2:

$$W = 2F\Delta x = \frac{QvBL}{4}(\cos\theta_2 - \cos\theta_1) = \mu B(\cos\theta_2 - \cos\theta_1).$$

This can be written as:

$$W = \vec{\mu} \cdot \vec{B}\big|_{\theta_2} - \vec{\mu} \cdot \vec{B}\big|_{\theta_1} = -\Delta U,$$

where $\Delta U = U(\theta_2) - U(\theta_1)$.

46.2 Potential Energy

From $W = -\Delta U$ (conservation of energy), we obtain:

$$U = -\vec{\mu} \cdot \vec{B}.$$

This is the potential energy of any current distribution (of arbitrary shape) in an external magnetic field.

46.3 Gyroscope vs. Pendulum Analogy

- A **spinning charged sphere** in a magnetic field behaves like a **gyroscope**: $\vec{\mu}$ precesses around $\vec{B}$ (due to angular momentum from spinning).
- A **current-carrying wire** in a magnetic field behaves like a **pendulum**: $\vec{\mu}$ swings back and forth (negligible angular momentum from the tiny electrons).

47 From the Spinning Sphere to the Ring and Sphere

47.1 Spinning Ring of Charge

For a ring of mass m, charge Q, radius R, spinning with rim velocity v: L = mvR,
I = Qv$\overline{2\pi R, A = \pi R^2.}$

The magnetic moment:

$$\mu = IA = \frac{QvR}{2} = \frac{Q}{2m} \cdot mvR = \frac{Q}{2m} L.$$

As a vector equation:

$$\vec{\mu} = \frac{Q}{2m} \vec{L}.$$

47.2 Spinning Sphere of Charge

A spinning sphere is an infinite stack of spinning rings. If the charge-to-mass ratio Q/m is constant throughout the volume, then $\vec{\mu} = \frac{Q}{2m}\vec{L}$ holds for the sphere as well (since both $\vec{L}$ and $\vec{\mu}$ are additive).

Therefore, the energy of a spinning charged sphere in an external magnetic field is:

$$U = -\frac{Q}{2m} \vec{L} \cdot \vec{B}.$$

48 Motivating the Schrödinger Equation

48.1 Quantum Time Evolution

A quantum spin state $\psi(t) = (\, c\,)_1\,(t)$
$c_2(t)$ evolves according to the **Schrödinger equation**:

$$i\hbar \frac{d}{dt} \psi(t) = H\psi(t),$$

where H is a 2×2 matrix called the **Hamiltonian**.

48.2 Dimensional Analysis

- i is dimensionless.
- $\hbar$ has dimensions of energy $\times$ time (equivalently, angular momentum).
- $\frac{d}{dt}\psi$ has dimensions of $1/time$ (since ψ is dimensionless).
- Therefore H must have dimensions of **energy**.

48.3 Constructing H: An Educated Guess

We take the classical energy $U = -\frac{Q}{2m}\vec{L}\cdot\vec{B}$ and "quantize" it:

[label=**Step 3:**]

1. **Charge:** For an electron, $Q = -e$, so $-Q/(2m) = e/(2m)$.
2. **Angular momentum:** In quantum mechanics, the spin angular momentum is $\vec{L} \to \frac{\hbar}{2}\vec{\sigma}$, where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices.
3. **Dot product:** $\vec{L}\cdot\vec{B} \to \frac{\hbar}{2}(B_x\sigma_x + B_y\sigma_y + B_z\sigma_z) = \frac{\hbar}{2}\vec{B}\cdot\vec{\sigma}$.

This yields:

$$H = \frac{ge}{2m}\cdot\frac{\hbar}{2}\big(B_x\sigma_x + B_y\sigma_y + B_z\sigma_z\big) = \frac{ge\hbar}{4m}\,\vec{B}\cdot\vec{\sigma},$$

where $g \approx 2$ is the **gyromagnetic ratio** (g-factor).

Remark 48.1 *The fact that $\hbar$ has units of both energy$\times$time* and *angular momentum is not a coincidence — it connects the Schrödinger equation (time evolution) with spin angular momentum.*

49 The Gyromagnetic Ratio and QFT

49.1 Why $g \approx 2$?

Classically, one would expect $g = 1$. The factor of $g \approx 2$ was first explained by **Paul Dirac** (1928) through his relativistic wave equation for the electron, which correctly accounts for special relativity.

49.2 Corrections from Quantum Field Theory

The g-factor is not exactly 2 but slightly larger. The first correction was computed by **Julian Schwinger** (1947):

$$g = 2 + \frac{\alpha}{\pi} + \mathcal{O}(\alpha^2),$$

where α is the **fine-structure constant**:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.036}.$$

The fine-structure constant provides a dimensionless measure of electromagnetic coupling strength.

49.3 Feynman Diagrams

In quantum field theory, the g-factor is computed by summing over **Feynman diagrams** — pictorial representations of particle interactions:

- The simplest diagram (one vertex) gives $g = 2$ (Dirac's result).
- The next diagram (one internal photon loop) contributes $+\alpha/\pi$ (Schwinger's correction).
- Higher-order diagrams contribute terms of order $\alpha^2, \alpha^3, \ldots$

Since $\alpha \approx 1/137 \ll 1$, higher-order corrections are increasingly small. The g-factor has been experimentally verified to **15 decimal places**:

$$g \approx 2.002\,319\,304\,361\,82\ldots$$

49.4 The Four Realms of Physics

	Slow ($v \ll c$)	**Fast** ($v \sim c$)
Big	Newtonian Mechanics	Special/General Relativity
Small	Quantum Mechanics (QM)	Quantum Field Theory (QFT)

The Schrödinger equation belongs to QM. It can be *motivated* from classical mechanics but not rigorously *derived* from it. A rigorous derivation requires QFT (the deeper theory). The Schrödinger equation is to QM what $F = ma$ is to Newtonian mechanics: a postulate justified by experimental success.

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Quantum Spin (5): Schrödinger's Equation & Matrix Exponentiation Lecture Notes on Quantum Spin

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50 Setup and Review

A point-like electron with spin state $\psi(t) = (\,c\,)_1\,(t)$
$c_2(t)$ sits in an external magnetic field $\vec{B}$. The Schrödinger equation governing time evolution is:

$$i\hbar \frac{d}{dt}\psi(t) = H\psi(t),$$

equivalently:

$$\frac{d}{dt}\psi(t) = -\frac{i}{\hbar}H\psi(t).$$

The Hamiltonian is $H = \mathcal{E}\,(\hat{n}\cdot\vec{\sigma})$, where:

- $\vec{B} = B\hat{n}$ ($\hat{n}$ is a unit vector, $B = |\vec{B}|$),
- $\mathcal{E} = \frac{ge\hbar B}{4m}$ is a constant with dimensions of energy,
- $\hat{n}\cdot\vec{\sigma} = n_x\sigma_x + n_y\sigma_y + n_z\sigma_z$ is a 2×2 matrix.

51 Method 1: Direct Solution

51.1 Example: $\vec{B}$ Along the y-Axis

Let $\hat{n} = \hat{y}$, so $H = \mathcal{E}\,\sigma_y = (\,0\,) - i\mathcal{E}$
$i\mathcal{E}0$.

Initial condition: $\psi(0) = (\,1\,)$
0 (spin up along z).

The Schrödinger equation becomes two coupled ODEs: $\dot{c}_1(t) = -\frac{\mathcal{E}}{\hbar}\,c_2(t)$, $\dot{c}_2(t) = +\frac{\mathcal{E}}{\hbar}\,c_1(t)$.

51.2 Decoupling

Differentiate eq:ode1 and substitute eq:ode2:

$$\ddot{c}_1 = -\frac{\mathcal{E}}{\hbar}\dot{c}_2 = -\frac{\mathcal{E}^2}{\hbar^2}\,c_1.$$

Similarly, $\ddot{c}_2 = -\frac{\mathcal{E}^2}{\hbar^2}\,c_2$. These are simple harmonic oscillator equations with angular frequency $\omega = \mathcal{E}/\hbar$.

51.3 General Solution and Initial Conditions

The general solutions are: $\mathrm{c}_1(t) = A\cos\!\left(\frac{\mathcal{E}}{\hbar}t\right) + B\sin\!\left(\frac{\mathcal{E}}{\hbar}t\right)$,
$c_2(t) = C\cos\!\left(\frac{\mathcal{E}}{\hbar}t\right) + D\sin\!\left(\frac{\mathcal{E}}{\hbar}t\right)$.

Using initial conditions $c_1(0) = 1$, $c_2(0) = 0$ and the original ODEs (guess and check):

$$\psi(t) = (\,\cos\,)(\mathcal{E}\hbar t)\sin(\mathcal{E}\hbar t)\,.$$

51.4 Verification

- $|c_1|^2 + |c_2|^2 = \cos^2(\mathcal{E}t/\hbar) + \sin^2(\mathcal{E}t/\hbar) = 1$ for all t.
- $\dot{c}_1 = -(\mathcal{E}/\hbar)\sin(\mathcal{E}t/\hbar) = -(\mathcal{E}/\hbar)c_2$.
- $\dot{c}_2 = +(\mathcal{E}/\hbar)\cos(\mathcal{E}t/\hbar) = +(\mathcal{E}/\hbar)c_1$.

51.5 Physical Interpretation: Larmor Precession

Evaluating $\psi(t)$ at key times (with $T_0 = \hbar/\mathcal{E}$):

t	$\psi(t)$	Direction
0	$(1,\ 0)^\top$	$+z$
$\frac{\pi}{4}T_0$	$\frac{1}{\sqrt{2}}(1,\ 1)^\top$	$+x$
$\frac{\pi}{2}T_0$	$(0,\ 1)^\top$	$-z$
$\frac{3\pi}{4}T_0$	$\frac{1}{\sqrt{2}}(-1,\ 1)^\top$	$-x$ (up to phase)
πT_0	$(-1,\ 0)^\top$	$+z$ (negated)

The Bloch vector traces a circle in the xz-plane, precessing around the y-axis (the direction of $\vec{B}$). After a full 2π rotation, $\psi \to -\psi$ — the state is *negated* (a signature of spin-$\frac{1}{2}$).

52 Method 2: Eigendecomposition

52.1 Finding Eigenstates of H

Since $H = \mathcal{E}\,\sigma_y$, the eigenstates of H are those of σ_y: H$\uparrow_y = +\mathcal{E}\uparrow_y$, $H\downarrow_y = -\mathcal{E}\downarrow_y$.

52.2 Eigenstates Yield Immediate Solutions

If $H\phi = E\phi$, then:

$$\phi(t) = e^{-iEt/\hbar}\phi$$

solves the Schrödinger equation. (Verification: $i\hbar\frac{d}{dt}\phi(t) = E\,e^{-iEt/\hbar}\phi = H\phi(t)$.)

The two solutions are: $\psi_1(t) = e^{-i\mathcal{E}t/\hbar}\uparrow_y$, $\psi_2(t) = e^{+i\mathcal{E}t/\hbar}\downarrow_y$.

52.3 Linearity and Superposition

Because the Schrödinger equation is **linear**, any linear combination of solutions is also a solution:

$$\psi(t) = \alpha\psi_1(t) + \beta\psi_2(t).$$

To match the initial condition $\psi(0) = (\,1\,)$
0:

$$(\,1\,)\,0 = \alpha\uparrow_y + \beta\downarrow_y \alpha = \beta = \frac{1}{\sqrt{2}}.$$

Therefore:

$$\psi(t) = \frac{1}{\sqrt{2}} e^{-i\mathcal{E}t/\hbar}\uparrow_y + \frac{1}{\sqrt{2}} e^{+i\mathcal{E}t/\hbar}\downarrow_y.$$

Expanding and using Euler's formula yields the same answer as Method 1:

$$\psi(t) = (\,\cos\,)\,(\mathcal{E}t/\hbar)\sin(\mathcal{E}t/\hbar).$$

53 Method 3: Matrix Exponentiation

53.1 Motivation via Time Stepping

From a first-order Taylor expansion:

$$\psi(t+\delta t) \approx \psi(t) + \delta t\,\frac{d}{dt}\psi(t) = \left(I - \frac{i}{\hbar}\delta t\,H\right)\psi(t).$$

Breaking time t into n small steps of size t/n:

$$\psi(t) \approx \left(I - \frac{i}{\hbar}\frac{t}{n}H\right)^n \psi(0).$$

Taking $n \to \infty$ and using the identity $e^A = \lim_{n\to\infty}\left(I + \frac{A}{n}\right)^n$:

$$\psi(t) = e^{-iHt/\hbar}\psi(0).$$

53.2 Rigorous Definition of the Matrix Exponential

Definition 53.1 (Matrix Exponential) *For a square matrix A:*

$$e^A \sum_{n=0}^{\infty} \frac{A^n}{n!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots$$

where $A^0 = I$ (the identity matrix).

53.3 Key Property

Theorem 53.1 (Differentiation of the Matrix Exponential)

$$\frac{d}{dt}e^{tA} = A\,e^{tA}.$$

$$\frac{d}{dt}e^{tA} = \frac{d}{dt}\sum_{n=0}^{\infty}\frac{t^n A^n}{n!} = \sum_{n=1}^{\infty}\frac{n\,t^{n-1}A^n}{n!} = A\sum_{n=1}^{\infty}\frac{t^{n-1}A^{n-1}}{(n-1)!} = A\,e^{tA}.$$

This immediately implies $\psi(t) = e^{-iHt/\hbar}\psi(0)$ solves the Schrödinger equation.

53.4 Computing Matrix Exponentials

53.4.1 Method A: Diagonalization

If $A = MDM^{-1}$ where D is diagonal, then $A^n = MD^nM^{-1}$ and:

$$e^A = M\,e^D\,M^{-1}, \quad where e^D = diag(e^{\lambda_1}, e^{\lambda_2}, \ldots).$$

53.4.2 Method B: Using $A^2 = -I$ (Generalized Euler Formula)

If A is a matrix satisfying $A^2 = -I$, then:

$$e^{\theta A} = \cos\theta\, I + \sin\theta\, A.$$

Separate the Taylor series into even and odd powers:

$$e^{\theta A} = \sum_{n=0}^{\infty} \frac{(\theta A)^n}{n!} = \underbrace{\sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k}}{(2k)!}}_{=\cos\theta} I + \underbrace{\sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k+1}}{(2k+1)!}}_{=\sin\theta} A,$$

using $A^{2k} = (-I)^k = (-1)^k I$ and $A^{2k+1} = (-1)^k A$.

Remark 53.1 *This is the matrix analog of Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$, since $i^2 = -1$ parallels $A^2 = -I$.*

54 Applying Matrix Exponentiation to Spin

54.1 The Time Evolution Operator

We need to compute $e^{-iHt/\hbar}$ where $H = \mathcal{E}(\hat{n}\cdot\vec{\sigma})$.

Define $A = -i(\hat{n}\cdot\vec{\sigma})$. Then $-iHt/\hbar = (\mathcal{E}t/\hbar)\,A$.

54.2 Verifying $A^2 = -I$

A$^2 = (-i)^2(\hat{n}\cdot\vec{\sigma})^2 = -(\hat{n}\cdot\vec{\sigma})^2$.

Computing $(\hat{n}\cdot\vec{\sigma})^2$ explicitly (with $|\hat{n}| = 1$):

$$(\hat{n}\cdot\vec{\sigma})^2 = (\,n\,)_z\, n_x - in_y n_x + in_y - n_z^2 = (\,n\,)_x^2 + n_y^2 + n_z^2 00 n_x^2 + n_y^2 + n_z^2 = I.$$

Therefore $A^2 = -I$.

54.3 The General Formula

Applying the generalized Euler formula:

$$e^{-iHt/\hbar} = \cos\left(\frac{\mathcal{E}t}{\hbar}\right) I + \sin\left(\frac{\mathcal{E}t}{\hbar}\right)\left(-i\,\hat{n}\cdot\vec{\sigma}\right).$$

54.4 Verification with the y-Axis Example

For $\hat{n} = \hat{y}$: $-i\sigma_y = -i\,(\,0\,) - i$
$i0 = (\,0\,) - 1$
10.

$$e^{-iHt/\hbar} = (\,\cos\,)\,(\mathcal{E}t/\hbar) - \sin(\mathcal{E}t/\hbar)\sin(\mathcal{E}t/\hbar)\cos(\mathcal{E}t/\hbar).$$

Acting on $\psi(0) = (\,1\,)$
0:

$$\psi(t) = (\,\cos\,)\,(\mathcal{E}t/\hbar)\sin(\mathcal{E}t/\hbar).$$

55 Time Evolution as Rotation

55.1 $\vec{B}$ Along the z-Axis

If $\hat{n} = \hat{z}$, then $H = \mathcal{E}\,\sigma_z$ and:

$$e^{-iHt/\hbar} = (\,e\,)^{-i\mathcal{E}t/\hbar}\,00e^{+i\mathcal{E}t/\hbar}.$$

Acting on a general state $\psi = (\,\cos\,)\,\frac{\theta}{2}$
$e^{i\phi}\sin\frac{\theta}{2}$:

$$\psi(t) = (\,e\,)^{-i\mathcal{E}t/\hbar}\cos\frac{\theta}{2}e^{i(\phi+2\mathcal{E}t/\hbar)}\sin\frac{\theta}{2} \sim (\,\cos\,)\,\frac{\theta}{2}e^{i(\phi+2\mathcal{E}t/\hbar)}\sin\frac{\theta}{2},$$

where $\sim$ denotes equality up to a global phase. The polar angle θ is unchanged; the azimuthal angle increases as $\phi \to \phi + 2\mathcal{E}t/\hbar$. The Bloch vector **rotates around the z-axis**.

55.2 General Rotation Formula

Theorem 55.1 (Spin Rotation) *The matrix*

$$U(\hat{n}, \varphi) = e^{-i\varphi\,\hat{n}\cdot\vec{\sigma}/2}$$

implements a ***rotation of the Bloch vector*** *around the axis $\hat{n}$ by angle φ.*

55.3 Two Key Takeaways

1. **Time evolution in a magnetic field *is* a rotation.** The Bloch vector precesses around $\hat{n}$ (the direction of $\vec{B}$) at a rate determined by $\mathcal{E}/\hbar$.

2. **The matrix exponential gives an explicit rotation formula.** Once $e^{-iHt/\hbar}$ is computed, applying it to *any* initial state immediately yields the time-evolved state.

55.4 Connection to Quantum Computing

In the language of quantum computation:

- The time evolution operator $U(t) = e^{-iHt/\hbar}$ is a **quantum gate**.
- A spin state is a **qubit**.
- Any computational operation on a qubit can be realized by choosing an appropriate magnetic field direction $\hat{n}$ and duration t.

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56 The Adjoint of a Matrix

Definition 56.1 (Adjoint (Conjugate Transpose)) *If A is an $m \times n$ matrix, its **adjoint** (or **conjugate transpose**) $A^\dagger$ is the $n \times m$ matrix with entries:*

$$(A^\dagger)_{ij} = \overline{A_{ji}}.$$

Equivalently, $A^\dagger$ is obtained by transposing A and complex conjugating every entry.

Example 56.1 *For a 2×2 matrix:*

$$A = (\,a\,)\,bcdA^\dagger = (^-)\,a\bar{c}\bar{b}\bar{d}.$$

Example 56.2 (Adjoint of a Ket) *For a column vector (ket):*

$$\psi = (\,\alpha\,)\,\beta\psi^\dagger = \psi = (^-)\,\alpha\bar{\beta}.$$

The adjoint of a ket is its corresponding bra (dual vector).

56.1 Key Property: Adjoint of a Product

Theorem 56.1 *For matrices A ($\ell \times m$) and B ($m \times n$):*

$$(AB)^\dagger = B^\dagger A^\dagger.$$

The order of multiplication reverses under the adjoint.

The (k, i)-entry of $B^\dagger A^\dagger$ is:

$$(B^\dagger A^\dagger)_{ki} = \sum_{j=1}^{m} (B^\dagger)_{kj}(A^\dagger)_{ji} = \sum_{j=1}^{m} \overline{B_{jk}}\,\overline{A_{ij}} = \overline{\sum_{j=1}^{m} A_{ij}B_{jk}} = \overline{(AB)_{ik}} = \big((AB)^\dagger\big)_{ki}.$$

57 The Pauli Matrices Are Self-Adjoint

Definition 57.1 (Self-Adjoint (Hermitian)) *A matrix A is **self-adjoint** (or **Hermitian**) if $A^\dagger = A$.*

Proposition 57.1 *Each Pauli matrix is self-adjoint: $\sigma_x^\dagger = \sigma_x$, $\sigma_y^\dagger = \sigma_y$, $\sigma_z^\dagger = \sigma_z$.*

Direct computation. For σ_y: transposing swaps the off-diagonal entries ($-i \leftrightarrow i$), and conjugating negates i back, restoring the original matrix. The others are even simpler since their entries are real.

58 The Hamiltonian H Is Self-Adjoint

Recall:

$$H = \frac{ge\hbar}{4m}\big(B_x\sigma_x + B_y\sigma_y + B_z\sigma_z\big) = \frac{ge\hbar}{4m}\,\vec{B}\cdot\vec{\sigma}.$$

Since $B_x, B_y, B_z \in R$ and $g, e, \hbar, m \in R$, and each σ_a is self-adjoint:

$$H^\dagger = \frac{ge\hbar}{4m}\big(B_x\sigma_x^\dagger + B_y\sigma_y^\dagger + B_z\sigma_z^\dagger\big) = \frac{ge\hbar}{4m}\big(B_x\sigma_x + B_y\sigma_y + B_z\sigma_z\big) = H.$$

Theorem 58.1 *The Hamiltonian H for a spin-$\frac{1}{2}$ particle in an external magnetic field is self-adjoint: $H^\dagger = H$.*

59 Consequence 1: Conservation of Probability

59.1 Statement

Theorem 59.1 (Conservation of Probability) *If $\psi(t)$ evolves according to the Schrödinger equation with a self-adjoint Hamiltonian H, then $\psi(t)\psi(t)$ is constant in time.*

In particular, if $\psi(t_0)\psi(t_0) = 1$ at some initial time t_0, then $\psi(t)\psi(t) = 1$ for all t. Probabilities continue to sum to 1 forever.

59.2 Proof

Compute the time derivative using the product rule:

$$\frac{d}{dt}\psi(t)\psi(t) = \left(\frac{d}{dt}\psi(t)\right)\psi(t) + \psi(t)\left(\frac{d}{dt}\psi(t)\right).$$

From the Schrödinger equation:

$$\frac{d}{dt}\psi = -\frac{i}{\hbar}H\psi.$$

Taking the adjoint of both sides yields the **dual Schrödinger equation**:

$$\frac{d}{dt}\psi = +\frac{i}{\hbar}\psi H^\dagger.$$

Substituting: $\mathrm{d}\overline{dt\psi\psi = \frac{i}{\hbar}\psi H^\dagger\psi - \frac{i}{\hbar}\psi H\psi = \frac{i}{\hbar}\psi\Big(H^\dagger - H\Big)\psi.}$

Since $H^\dagger = H$:

$$\frac{d}{dt}\psi\psi = \frac{i}{\hbar}\psi\cdot\mathbf{0}\cdot\psi = 0.$$

Remark 59.1 *The proof actually shows something stronger: $\psi(t)\psi(t)$ is constant for* any *initial value, not just* 1*. Even an "unphysical" state with $\psi\psi = 5$ would retain that value under Schrödinger evolution.*

60 Consequence 2: Uniqueness of Time Evolution

60.1 The Question

If two solutions $\psi_1(t)$ and $\psi_2(t)$ of the Schrödinger equation agree at some time t_0:

$$\psi_1(t_0) = \psi_2(t_0),$$

must they agree at *all* times?

60.2 Why Naïve Taylor Expansion Fails

One might try: expand both solutions to first order, note they agree at t_0, so they agree at $t_0 + \delta t$, then "inductively" extend. However, this argument is **invalid** — you cannot extend an inductive argument through an uncountable infinity of infinitesimal steps.

Example 60.1 (A Counterexample to Naïve Reasoning) *The ODE $\dot{y} = y^{2/3}$ with $y(0) = 0$ has infinitely many solutions:*

$$y(t) = \{\, 0 \; t \leq t^*, \frac{1}{27}(t - t^*)^3 t > t^*,$$

*for any $t^* > 0$. All agree at $t = 0$ (and to all Taylor orders), yet diverge later. This ODE violates the **Lipschitz condition**, which guarantees uniqueness.*

60.3 The Correct Proof

Theorem 60.1 (Uniqueness) *If $\psi_1(t_0) = \psi_2(t_0)$ and both solve the Schrödinger equation, then $\psi_1(t) = \psi_2(t)$ for all t.*

Define $\tilde{\psi}(t)\psi_1(t) - \psi_2(t)$.

Step 1: $\tilde{\psi}(t)$ solves the Schrödinger equation (by linearity).

Step 2: At time t_0: $\tilde{\psi}(t_0) = 0$, so $\tilde{\psi}(t_0)\tilde{\psi}(t_0) = 0$.

Step 3: By conservation of the inner product (Consequence 1): $\tilde{\psi}(t)\tilde{\psi}(t) = 0$ for all t.

Step 4: Since $\tilde{\psi}\tilde{\psi} = |\tilde{\alpha}|^2 + |\tilde{\beta}|^2 = 0$ implies $\tilde{\alpha} = \tilde{\beta} = 0$, we conclude $\tilde{\psi}(t) = 0$ for all t.

Therefore $\psi_1(t) = \psi_2(t)$ for all t.

Remark 60.1 *The proof crucially uses conservation of the inner product, which itself relies on H being self-adjoint. Even without self-adjointness, uniqueness still holds (by more general ODE theory), but the proof is much harder.*

61 Consequence 3: Eigenvalues of H Are Real

61.1 Statement and Proof

Theorem 61.1 *If H is self-adjoint, then every eigenvalue of H is a real number.*

Let $H\psi = E\psi$ with $\psi \neq 0$.
Taking the adjoint: $\psi H^\dagger = \bar{E}\psi$, i.e., $\psi H = \bar{E}\psi$ (since $H^\dagger = H$).
Now compute $\psi H \psi$ in two ways: $\psi\big(H\psi\big) = E\psi\psi$, $\big(\psi H\big)\psi = \bar{E}\psi\psi$.
Since $\psi\psi > 0$ (as $\psi \neq 0$), divide both sides:

$$E = \bar{E}E \in R.$$

61.2 Physical Interpretation: Energy Is Real

In quantum mechanics, the eigenvalues of H are the **possible energies** a state can have:

- If $H\psi = E\psi$, then ψ is a state with **definite energy** E.
- A superposition $\psi = \alpha\psi_1 + \beta\psi_2$ (where $H\psi_k = E_k\psi_k$) does *not* have a definite energy — measuring the energy yields E_1 with probability $|\alpha|^2$ and E_2 with probability $|\beta|^2$.

The theorem guarantees that all measurable energies are real numbers — quantum mechanics is strange, but not so strange that you can measure an imaginary energy.

Remark 61.1 *The statement "eigenvalues of H are the energies" is an **interpretive postulate** of quantum mechanics, not a theorem. It is the quantum-mechanical* definition *of energy. In appropriate limits, it reduces to the classical notion of energy.*

62 Real Eigenvalues $\Longleftrightarrow$ Probability Conservation

These two consequences are intimately related. For an energy eigenstate ψ with eigenvalue E:

$$\psi(t) = e^{-iEt/\hbar}\psi(0).$$

$$\psi(t)\psi(t) = e^{i(\bar{E}-E)t/\hbar}\psi(0)\psi(0).$$

- If $E \in R$: $\bar{E} - E = 0$, so $\psi(t)\psi(t) = \psi(0)\psi(0)$. Probability is conserved.
- If $E = a + ib$ with $b \neq 0$: $\psi(t)\psi(t) = e^{2bt/\hbar}\psi(0)\psi(0)$, which grows or decays exponentially — probability is *not* conserved. $\times$

Thus: $E \in R$ (real energies) $\Longleftrightarrow$ probability conservation.

63 Consequence 4: Time Evolution Is Unitary

The time evolution operator is $U(t) = e^{-iHt/\hbar}$.

Definition 63.1 (Unitary Matrix) *A matrix U is **unitary** if $U^\dagger U = I$.*

Theorem 63.1 *If H is self-adjoint, then $U(t) = e^{-iHt/\hbar}$ is unitary for all t.*

Compute:

$$U(t)^\dagger = \left(e^{-iHt/\hbar}\right)^\dagger = e^{+iH^\dagger t/\hbar} = e^{+iHt/\hbar},$$

using $H^\dagger = H$. Then:

$$U(t)^\dagger U(t) = e^{+iHt/\hbar} e^{-iHt/\hbar} = e^{i(H-H)t/\hbar} = e^0 = I.$$

Remark 63.1 *Unitarity of $U(t)$ is the matrix-level statement that inner products are preserved under time evolution: $U\psi U\psi = \psi U^\dagger U\psi = \psi\psi$.*

63.1 Summary of the Logical Chain

$H = H^\dagger$ { $ProbabilityisconservedEigenvalues(energies)arerealTimeevolutionisuniqueTimeevolutionisunit$

All four consequences flow from one mathematical property: the self-adjointness of the Hamiltonian.

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Quantum Spin (7): Wavefunctions & Motion in a Magnetic Field Lecture Notes on Quantum Spin

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64 Particle Motion Without Spin

Up to this point, we have treated the electron as a stationary object with a spin state $\psi \in C^2$. We now address how electrons *move* through space.

64.1 The Wavefunction

Definition 64.1 (Wavefunction) *The state of a spinless quantum particle is described by a* ***wavefunction*** $\psi(t, \vec{x})$*, where* $t \in R$ *is time and* $\vec{x} = (x, y, z) \in R^3$ *is position. The wavefunction outputs a complex number:*

$$\psi : R \times R^3 \to C.$$

Definition 64.2 (Probability Density) *The quantity* $|\psi(t, \vec{x})|^2$ *is the* ***probability density*** *of finding the particle at position* $\vec{x}$ *at time* t*. Concretely, the probability of finding the particle in an infinitesimal volume* dV *around* $\vec{x}$ *is:*

$$P(\vec{x} \in dV) = |\psi(t, \vec{x})|^2 \, dV.$$

64.2 Normalization

The total probability of finding the particle *somewhere* in all of space must be 1:

$$\int_{R^3} |\psi(t, \vec{x})|^2 \, d^3x = 1 \quad forallt.$$

64.3 The Free-Particle Schrödinger Equation

The wavefunction evolves according to the **Schrödinger equation** (for a free particle, no external potential):

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi,$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the Laplacian and m is the particle mass.

64.4 Connection to Classical Kinetic Energy

Classically, kinetic energy is $T = \frac{p^2}{2m} = \frac{p_x^2 + p_y^2 + p_z^2}{2m}$. In quantum mechanics, momentum becomes a differential operator:

$$\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}, \qquad \hat{p}_x^2 = -\hbar^2 \frac{\partial^2}{\partial x^2}.$$

The right-hand side of the Schrödinger equation is thus the quantum kinetic energy operator acting on ψ.

65 Gaussian Wave Packets

65.1 An Important Solution

A particularly important solution to the free-particle Schrödinger equation is the **Gaussian wave packet**:

$$\psi(t,\vec{x}) = \frac{\pi^{-3/4}\,\sigma^{-3/2}}{c(t)^{3/2}} \exp\left[-\frac{|\vec{x}|^2}{2\sigma^2 c(t)} + \frac{i\,\vec{p}\cdot\vec{x}}{\hbar} - \frac{i\,|\vec{p}|^2 t}{2m\hbar\,c(t)}\right],$$

where:

- $\sigma > 0$ is the **initial position uncertainty** (width at $t = 0$),
- $\vec{p} = (p_x, p_y, p_z) \in R^3$ is the **momentum** (determines the velocity),
- $c(t) = 1 + \frac{i\hbar t}{m\sigma^2}$ is a time-dependent complex factor.

65.2 Probability Density

The probability density $|\psi(t,\vec{x})|^2$ is a Gaussian (bell curve) centered at $\vec{x}_{peak} = \frac{\vec{p}\,t}{m}$:

$$|\psi(t,\vec{x})|^2 \propto \exp\left[-\frac{|\vec{x} - \vec{p}t/m|^2}{\sigma^2 + (\hbar t/m\sigma)^2}\right].$$

Key features:

1. **Center:** The peak moves at the classical velocity $\vec{v} = \vec{p}/m$.
2. **Width:** The spatial spread at time t is $\Delta x(t) = \sqrt{\sigma^2 + \left(\frac{\hbar t}{m\sigma}\right)^2}$.
3. At $t = 0$: $\Delta x(0) = \sigma$ (the initial uncertainty).
4. As $t \to \infty$: $\Delta x \approx \frac{\hbar t}{m\sigma}$ (the wave packet spreads).

65.3 Real and Imaginary Parts

While the envelope $|\psi|$ moves smoothly to the right and broadens, the real and imaginary parts of ψ oscillate rapidly. The oscillation frequency encodes the velocity: higher velocity $\Rightarrow$ faster oscillation.

66 Particle Motion with Spin

66.1 Two-Component Wavefunctions

When a particle has both spatial degrees of freedom *and* spin, the state is described by a **two-component wavefunction** (a spinor-valued wavefunction):

$$\Psi(t,\vec{x}) = (\,\psi\,)_{\uparrow}\,(t,\vec{x})\psi_{\downarrow}(t,\vec{x}),$$

where $\psi_{\uparrow}$ and $\psi_{\downarrow}$ are each complex-valued functions of $(t,\vec{x})$.

66.2 Interpretation

- $|\psi_\uparrow(t,\vec{x})|^2$: probability density of finding the particle at $\vec{x}$, at time t, with spin **up** along z.
- $|\psi_\downarrow(t,\vec{x})|^2$: probability density of finding the particle at $\vec{x}$, at time t, with spin **down** along z.

For other spin axes, form linear combinations. For example, the probability density of finding the particle at $\vec{x}$ with spin up along x:

$$P(\vec{x},\uparrow_x) = \left|\frac{1}{\sqrt{2}}\psi_\uparrow(t,\vec{x}) + \frac{1}{\sqrt{2}}\psi_\downarrow(t,\vec{x})\right|^2.$$

66.3 Normalization

$$\int_{R^3}\Big(|\psi_\uparrow(t,\vec{x})|^2 + |\psi_\downarrow(t,\vec{x})|^2\Big)\,d^3x = 1 \quad forallt.$$

67 The Pauli Equation

The time evolution of a spin-$\frac{1}{2}$ charged particle in an electromagnetic field is governed by the **Pauli equation**:

$$i\hbar\frac{\partial}{\partial t}\,(\,\psi\,)_\uparrow\,\psi_\downarrow = \left[\frac{1}{2m}\left(\frac{\hbar}{i}\vec{\nabla} - q\vec{A}\right)^2 I + q\Phi\, I + \frac{q}{2m}\hbar\,\vec{B}\cdot\vec{\sigma}\right](\,\psi\,)_\uparrow\,\psi_\downarrow,$$

where I is the 2×2 identity matrix.

67.1 Terms in the Pauli Equation

[label=**Term 1:**]

1. $\frac{1}{2m}\left(\frac{\hbar}{i}\vec{\nabla} - q\vec{A}\right)^2 I$: **Kinetic energy** including the magnetic vector potential $\vec{A}$ (where $\vec{B} = \nabla\times\vec{A}$). This term:
   - Governs the spatial motion of the particle.
   - Encodes the **Lorentz force**: $\vec{F} = q\vec{v}\times\vec{B}$.
   - Multiplies the identity matrix I (acts on both spin components equally).
2. $q\Phi\, I$: **Electric potential energy**, where $\vec{E} = -\nabla\Phi - \frac{\partial\vec{A}}{\partial t}$. Encodes the **electric force**: $\vec{F} = q\vec{E}$.
3. $\frac{q}{2m}\hbar\,\vec{B}\cdot\vec{\sigma}$: **Spin–magnetic field coupling** (with $g\approx 2$). This is the term we studied in previous lectures. It:

- Couples the two spin components $\psi_\uparrow$ and $\psi_\downarrow$.
- Causes spin precession (Larmor precession).
- Gives rise to the **force on a magnetic moment**: $\vec{F} = \nabla(\vec{\mu} \cdot \vec{B})$ when $\vec{B}$ varies in space.

Remark 67.1 *In the earlier lectures (Lectures 4–6), we studied only Term 3 in isolation, treating the electron as stationary. The Pauli equation is the full picture incorporating spatial motion.*

68 Force on a Magnetic Moment: $\vec{F} = \nabla(\vec{\mu} \cdot \vec{B})$

68.1 Qualitative Picture

When a current loop (or spinning charge) with magnetic moment $\vec{\mu}$ is placed in a *spatially varying* magnetic field $\vec{B}(\vec{x})$:

- If $\vec{B}$ is uniform: no net force (only torque $\Rightarrow$ precession).
- If $\vec{B}$ varies in space: there is a net force.

Consider a current loop in a field that is stronger at the bottom than the top. The field lines splay outward, creating lateral force components on the current. Using the Lorentz force $\vec{F} = q\vec{v} \times \vec{B}$ on each segment of the loop and summing, one finds a net downward force.

68.2 Derivation for a Square Current Loop

Consider a square loop with side L, current $I = Qv/(4L)$, area $A = L^2$, centered at the origin in the xy-plane, with $\vec{\mu} = IA\,\hat{z}$.

The magnetic field on each segment is found by Taylor expansion to first order in L: $\vec{B}\big|_{bottom} \approx \vec{B}_0 + \frac{L}{2}\partial_x\vec{B}, \vec{B}\big|_{top} \approx \vec{B}_0 - \frac{L}{2}\partial_x\vec{B}$, $\vec{B}\big|_{right} \approx \vec{B}_0 - \frac{L}{2}\partial_y\vec{B}, \vec{B}\big|_{left} \approx \vec{B}_0 + \frac{L}{2}\partial_y\vec{B}$.

Computing $\vec{F} = (Q/4)\vec{v} \times \vec{B}$ on each segment and summing:

$$\vec{F}_{net} = \frac{QvL}{4}\,(\,\partial\,)_x\, B_z\partial_y B_z - \partial_x B_x - \partial_y B_y.$$

Using Maxwell's equation $\nabla \cdot \vec{B} = 0$ (i.e., $\partial_x B_x + \partial_y B_y + \partial_z B_z = 0$):

$$-\partial_x B_x - \partial_y B_y = \partial_z B_z.$$

Since $\vec{\mu} \cdot \vec{B} = IAB_z$ and $IA = QvL/4$:

$$\vec{F}_{net} = \nabla(\vec{\mu} \cdot \vec{B}).$$

Remark 68.1 *This result crucially depends on $\nabla \cdot \vec{B} = 0$ (no magnetic monopoles). Without this Maxwell equation, the z-component would not simplify correctly.*

68.3 Generalization

The force law $\vec{F} = \nabla(\vec{\mu} \cdot \vec{B})$ is valid for:

- Any shape of current loop (decompose into tiny squares; internal currents cancel).
- A charged spinning sphere with uniform charge-to-mass ratio (decompose into spinning rings).
- A quantum spin-$\frac{1}{2}$ particle (via the Pauli equation).

68.4 Quick Derivation via Potential Energy

From earlier lectures: $U = -\vec{\mu} \cdot \vec{B}$. The force is:

$$\vec{F} = -\nabla U = \nabla(\vec{\mu} \cdot \vec{B}).$$

This recovers the same result in one line, though the direct Lorentz-force derivation provides more physical insight.

69 Classical Intuition for Quantum Wave Packets

For Gaussian wave packets, the average position evolves approximately according to classical mechanics:

- The **Lorentz force** $\vec{F} = q\vec{v} \times \vec{B}$ deflects the wave packet's trajectory (from Term 1).
- The **electric force** $\vec{F} = q\vec{E}$ accelerates the wave packet (from Term 2).
- The **magnetic moment force** $\vec{F} = \nabla(\vec{\mu} \cdot \vec{B})$ deflects the wave packet depending on its spin (from Term 3).

This classical-quantum correspondence (Ehrenfest's theorem) allows us to understand qualitatively how quantum particles move without solving the full Pauli equation.

70 Preview: The Stern–Gerlach Experiment

The magnetic moment force provides the mechanism for the **Stern–Gerlach experiment**:

1. A particle emitter sends a spin-$\frac{1}{2}$ particle (as a wave packet) into a region with a spatially varying magnetic field ($\partial B_z/\partial z \neq 0$).

2. The spin-up component $\psi_\uparrow$ feels an upward force; the spin-down component $\psi_\downarrow$ feels a downward force.

3. The two components spatially separate, hitting a detector screen at distinct locations.

4. This spatial separation constitutes a **measurement of spin**.

Classically, a spinning object could have angular momentum pointing in *any* direction, so one would expect a continuous distribution on the screen. Quantum mechanically, only two spots appear — direct evidence of spin quantization.

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Quantum Spin (8): The Stern–Gerlach Experiment, Measurement, & Entanglement Lecture Notes on Quantum Spin

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71 The Stern–Gerlach Experiment

71.1 Basic Setup

The **Stern–Gerlach experiment** provides a concrete, physical realization of a quantum spin measurement. The apparatus consists of:

1. **Particle emitter:** Emits spin-$\frac{1}{2}$ particles (historically, silver atoms) traveling horizontally.
2. **Inhomogeneous magnetic field:** A region where the z-component of $\vec{B}$ varies with height: $\frac{\partial B_z}{\partial z} \neq 0$.
3. **Detector screen (back wall):** Records where the particle strikes.

71.2 The Deflection Mechanism

From the previous lecture, the force on a magnetic moment in an inhomogeneous field is:

$$\vec{F} = \nabla(\vec{\mu} \cdot \vec{B}).$$

For a spin-$\frac{1}{2}$ particle whose spin is along $+z$ (spin up), $\vec{\mu} = \mu\,\hat{z}$:

$$F_z = \mu \frac{\partial B_z}{\partial z}.$$

For spin down, $\vec{\mu} = -\mu\,\hat{z}$:

$$F_z = -\mu \frac{\partial B_z}{\partial z}.$$

The spin-up component is deflected upward; the spin-down component is deflected downward.

71.3 Quantum vs. Classical Predictions

- **Classical prediction:** Angular momentum $\vec{L}$ can point in any direction $\Rightarrow$ a continuous smear on the detector screen.
- **Quantum prediction:** Spin is quantized (only up or down along the measurement axis) $\Rightarrow$ exactly **two discrete spots** on the screen.

The observation of two spots was among the first direct experimental evidence of spin quantization.

71.4 Quantum-Mechanical Description

The particle's state is a two-component wavefunction:

$$\Psi(t, \vec{x}) = (\,\psi\,)_{\uparrow}\,(t, \vec{x})\psi_{\downarrow}(t, \vec{x}).$$

Initially, both components overlap spatially (same Gaussian wave packet). As the particle traverses the inhomogeneous field:

- $\psi_{\uparrow}$ is deflected upward (feels $+F_z$).
- $\psi_{\downarrow}$ is deflected downward (feels $-F_z$).

After passing through the field, the two wave packets are **spatially separated**: $\psi_{\uparrow}$ is localized near the upper spot, $\psi_{\downarrow}$ near the lower spot.

72 Why Silver Atoms?

The original Stern–Gerlach experiment (1922) used **silver atoms**, not lone electrons. There are several important reasons:

72.1 Eliminating the Lorentz Force

An electron in a magnetic field experiences the Lorentz force $\vec{F} = q\vec{v} \times \vec{B}$, which would deflect it regardless of spin, overwhelming the much weaker magnetic-moment force. Silver atoms are **electrically neutral** ($Q_{total} = 0$), so the Lorentz force vanishes.

72.2 Silver's Electronic Structure

Silver (atomic number $Z = 47$) has one **valence electron** in the $5s$ orbital:

- All inner-shell electrons are paired (one spin up, one spin down per pair), so their magnetic moments cancel.
- The lone valence electron accounts for the net spin of the entire atom.
- Protons and neutrons also have spin, but $\mu \propto q/(2m)$ and the proton mass is $\sim 2000\times$ the electron mass, making nuclear magnetic moments negligible in comparison.
- The s-orbital has **zero orbital angular momentum** ($\ell = 0$), so only the intrinsic spin contributes — no complications from orbital motion.

72.3 Creating the Inhomogeneous Field

The varying magnetic field is produced by two permanent magnets:

- The **top magnet** has a *pointed* (wedge-shaped) pole face.
- The **bottom magnet** has a *flat* (rectangular) pole face.

Inside a ferromagnet, all atomic spins are aligned, producing dense, parallel field lines. Outside, the lines must loop back (since $\nabla \cdot \vec{B} = 0$, no magnetic monopoles). The pointed top pole causes field lines to splay outward more rapidly, creating a higher field density near the top than the bottom:

$$\frac{\partial B_z}{\partial z} \neq 0.$$

Two flat pole faces would produce a nearly uniform field ($\partial B_z/\partial z \approx 0$), insufficient for spin separation.

73 Lessons About Quantum Measurement

The Stern–Gerlach experiment illuminates four fundamental lessons about measurement in quantum mechanics.

73.1 Lesson 1: Measurements Take Time

Measurement is *not* instantaneous. The particle must travel from the emitter, through the magnetic field region, to the detector. The spatial separation of the spin components develops gradually over this transit time.

73.2 Lesson 2: Measurements Are Not Perfectly Accurate

Each spin component arrives at the screen as a Gaussian wave packet with finite width σ:

$$|\psi_\uparrow(z)|^2 \propto e^{-(z-z_\uparrow)^2/2\sigma^2}, \qquad |\psi_\downarrow(z)|^2 \propto e^{-(z-z_\downarrow)^2/2\sigma^2}.$$

The tails of the two Gaussians overlap slightly, introducing a small probability of misidentification. However, making the magnets *longer* increases the separation $|z_\uparrow - z_\downarrow|$, making the measurement arbitrarily precise (at the cost of taking longer).

73.3 Lesson 3: Measurement Is Not Mystical

Measurement requires a real physical apparatus — magnets, detectors, careful engineering. Pauli matrices and eigenvectors are mathematical abstractions; nobody has ever "seen" a Pauli matrix in the lab. The physics lies in how real devices exploit electromagnetic interactions to extract information about quantum states.

73.4 Lesson 4: Measurement Works via Entanglement

This is the deepest lesson and requires the concept of entanglement, developed below.

74 Hilbert Spaces and the Tensor Product

74.1 Hilbert Spaces for Different Degrees of Freedom

Definition 74.1 (Hilbert Space) *A **Hilbert space** $\mathcal{H}$ is a (complete) inner product space that serves as the space of states for a quantum system.*

- **Spin only:** $\mathcal{H}_{spin} = C^2$ (two-component complex vectors).
- **Position only:** $\mathcal{H}_{pos} = \{f : R^3 \to C\}$ (square-integrable wavefunctions).
- **Both:** $\mathcal{H} = \mathcal{H}_{pos} \otimes \mathcal{H}_{spin}$ (functions from R^3 to C^2).

74.2 The Tensor Product

Definition 74.2 (Tensor Product) *The **tensor product** $\otimes$ combines states from different Hilbert spaces into a joint state. If $\phi \in \mathcal{H}_1$ and $\chi \in \mathcal{H}_2$, then $\phi \otimes \chi \in \mathcal{H}_1 \otimes \mathcal{H}_2$.*

Key properties:

1. **Distributive:** $\phi \otimes \big(\alpha\chi_1 + \beta\chi_2\big) = \alpha\,\phi \otimes \chi_1 + \beta\,\phi \otimes \chi_2$.
2. **General states:** A general state in $\mathcal{H}_1 \otimes \mathcal{H}_2$ is a *sum* of tensor products, not necessarily a single tensor product.

In our context:

- $\psi_{init}(\vec{x}) \otimes \uparrow$: a wave packet ψ_{init} that is spin up.
- $\psi_{init}(\vec{x}) \otimes \downarrow$: the same wave packet, spin down.
- A general state with both spin and position:

$$\Psi = \psi_\uparrow(\vec{x}) \otimes \uparrow + \psi_\downarrow(\vec{x}) \otimes \downarrow \;\longleftrightarrow\; (\,\psi\,)_\uparrow(\vec{x})\psi_\downarrow(\vec{x}).$$

75 Entanglement

75.1 Definition

Definition 75.1 (Product State vs. Entangled State) *A state $\Psi \in \mathcal{H}_1 \otimes \mathcal{H}_2$ is a **product state** (not entangled) if it can be written as a single tensor product:*

$$\Psi = \phi \otimes \chi.$$

*It is **entangled** if it* cannot *be written in this form, i.e., it requires a sum of tensor products.*

75.2 Entanglement as Correlation

Entanglement is analogous to **correlation**: knowledge of one degree of freedom provides information about the other.

- **Product state** $\psi_{init}(\vec{x}) \otimes \big(\alpha\uparrow + \beta\downarrow\big)$: knowing the particle's position tells you nothing about its spin. The position and spin are *independent*.
- **Entangled state** $\alpha\,\psi_{up}(\vec{x}) \otimes \uparrow + \beta\,\psi_{down}(\vec{x}) \otimes \downarrow$ (with $\psi_{up} \not\propto \psi_{down}$): finding the particle at the "upper" position implies it is spin up; finding it at the "lower" position implies spin down. Position and spin are *correlated*.

76 The Stern–Gerlach Experiment as Entanglement

76.1 Before the Magnetic Field

The initial state is a product state:

$$\Psi_{init} = \psi_{init}(\vec{x}) \otimes \big(\alpha\uparrow + \beta\downarrow\big).$$

By the distributive property:

$$= \alpha\,\psi_{init}(\vec{x}) \otimes \uparrow + \beta\,\psi_{init}(\vec{x}) \otimes \downarrow.$$

Both terms share the *same* spatial wavefunction ψ_{init} — a product state. Position and spin are uncorrelated.

76.2 After the Magnetic Field

Time evolution via the Pauli equation (which is linear) maps each term independently: $\psi_{init}(\vec{x}) \otimes \uparrow$ *evolve* $\psi_{f,\uparrow}(\vec{x}) \otimes \uparrow$,
$\psi_{init}(\vec{x}) \otimes \downarrow$ *evolve* $\psi_{f,\downarrow}(\vec{x}) \otimes \downarrow$.

By linearity:

$$\Psi_{final} = \alpha\,\psi_{f,\uparrow}(\vec{x}) \otimes \uparrow + \beta\,\psi_{f,\downarrow}(\vec{x}) \otimes \downarrow.$$

Since $\psi_{f,\uparrow}$ and $\psi_{f,\downarrow}$ are spatially separated (different peaks), $\psi_{f,\uparrow} \not\propto \psi_{f,\downarrow}$, so this state is **entangled**. The measurement apparatus has entangled the spin degree of freedom with the position degree of freedom.

Theorem 76.1 (Measurement as Entanglement) *The role of a measurement apparatus is to take an initial product state (spin and position uncorrelated) and evolve it into an entangled state (spin and position correlated), so that observing the position reveals the spin.*

76.3 Degree of Entanglement

The quality of the measurement is related to how well-separated the spatial wavefunctions are. Consider:

$$|\psi_{f,\uparrow}\psi_{f,\downarrow}|^2 = \left| \int \overline{\psi_{f,\uparrow}(\vec{x})}\, \psi_{f,\downarrow}(\vec{x})\, d^3x \right|^2 .$$

- ≈ 0 (negligible overlap): **strong entanglement**, excellent measurement.
- ≈ 1 (large overlap): **weak entanglement**, poor measurement.

Making the magnets longer decreases the overlap, improving the measurement.

77 Quantum Measurement Collapse

77.1 The Puzzle

After the Stern–Gerlach apparatus, the state is the entangled superposition:

$$\Psi_{final} = \alpha\, \psi_{f,\uparrow} \otimes \uparrow + \beta\, \psi_{f,\downarrow} \otimes \downarrow.$$

This is a superposition of "particle at top, spin up" and "particle at bottom, spin down." Yet we never *observe* the superposition — we see the particle at one location *or* the other:

- With probability $|\alpha|^2$: particle found at the upper spot, spin up.
- With probability $|\beta|^2$: particle found at the lower spot, spin down.

This is the **measurement problem**: why does the superposition "collapse" to one outcome?

Remark 77.1 *This is essentially* ***Schrödinger's cat*** *in miniature: why do we see the cat alive* or *dead, but never in a superposition of both?*

77.2 Quantum Decoherence: A Partial Answer

The resolution involves the **environment**. The full Hilbert space is:

$$\mathcal{H} = \mathcal{H}_{pos} \otimes \mathcal{H}_{spin} \otimes \mathcal{H}_{env}.$$

When the particle hits the detector screen, it displaces atoms in the screen, which displace neighboring atoms, which affect air molecules, etc. The state becomes:

$$\alpha\, \psi_{f,\uparrow} \otimes \uparrow \otimes E_{f,\uparrow} + \beta\, \psi_{f,\downarrow} \otimes \downarrow \otimes E_{f,\downarrow},$$

where $E_{f,\uparrow}$ and $E_{f,\downarrow}$ are the environmental states corresponding to each outcome.

Proposition 77.1 *For a macroscopic environment with many degrees of freedom:*

$$|E_{f,\uparrow} E_{f,\downarrow}|^2 \, 1.$$

The environmental states are effectively orthogonal because two "random" states in a very high-dimensional Hilbert space have negligible overlap.

This **three-way entanglement** (position $\otimes$ spin $\otimes$ environment) is the mechanism of **quantum decoherence**: the environment "measures" the experiment, causing the superposition to behave *as if* it had collapsed to a single outcome.

Definition 77.1 (Quantum Decoherence) ***Decoherence*** *is the process by which a quantum system becomes entangled with its environment, causing quantum superpositions to become effectively classical (no observable interference between branches).*

Remark 77.2 *Decoherence explains* why we don't see superpositions in everyday life. *It does not, by itself, fully resolve the measurement problem (the question of which specific outcome is realized remains open and is the subject of various interpretations of quantum mechanics). For a thorough treatment, see M. Schlosshauer,* Quantum Decoherence *(Springer).*

78 Summary of the Quantum Spin Series

Over the course of these eight lectures, we have developed a complete picture of quantum spin for spin-$\frac{1}{2}$ particles:

1. **Lecture 1 (Introduction):** Spin states live in C^2; measurements yield probabilities via inner products; collapse occurs after measurement.

2. **Lecture 2 (Pauli Matrices):** The three Pauli matrices $\sigma_x, \sigma_y, \sigma_z$ compute expectation values and encode the spin observable along each axis.

3. **Lecture 3 (Bloch Sphere):** Every spin state corresponds to a point on the unit sphere via the Bloch vector $\hat{n} = \psi \vec{\sigma} \psi$.

4. **Lecture 4 (Classical Dynamics):** Classical Larmor precession motivates the quantum Hamiltonian $H = \mathcal{E}\, \hat{n} \cdot \vec{\sigma}$; the gyromagnetic ratio $g \approx 2$ arises from relativistic quantum field theory.

5. **Lecture 5 (Schrödinger's Equation):** Three methods for solving the Schrödinger equation; the matrix exponential $e^{-iHt/\hbar}$ implements time evolution as a rotation of the Bloch vector.

6. **Lecture 6 (Self-Adjointness):** $H = H^\dagger$ implies conservation of probability, uniqueness of time evolution, real energy eigenvalues, and unitarity.

7. **Lecture 7 (Wavefunctions):** The Pauli equation governs spatial motion with spin; the force $\vec{F} = \nabla(\vec{\mu} \cdot \vec{B})$ on a magnetic moment in an inhomogeneous field.

8. **Lecture 8 (Measurement):** The Stern–Gerlach experiment measures spin by entangling it with position; decoherence via environmental entanglement explains the apparent collapse of the wavefunction.

Remark 78.1 *The spin-$\frac{1}{2}$ system, despite being the simplest quantum system (C^2), contains essentially every conceptual feature of quantum mechanics: superposition, measurement, probability, collapse, time evolution, unitarity, entanglement, and decoherence. Mastery of this system provides a foundation for all of quantum mechanics.*